

Integrated Transcriptomic Analysis of PCOS reveals Metabolic and Endocrine-Transcriptional Axes

FINAL PROJECT REPORT



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Abstract

Polycystic ovary syndrome (PCOS) is a complex endocrine-metabolic disorder affecting up to 20% of reproductive-aged women, characterized by hormonal imbalance, insulin resistance, and chronic low-grade inflammation. While cortisol dysregulation has been implicated in PCOS pathophysiology, the specific transcriptional signatures linking stress response to ovarian dysfunction remain poorly defined. This study performed an integrative in silico analysis to map cortisol-associated gene networks across multiple PCOS transcriptomic datasets.

We analyzed four publicly available datasets: three microarray studies (GSE137684, GSE114419, GSE98421) and one RNA-seq dataset (GSE277906), encompassing granulosa cells, cumulus cells, endometrial, and adipose tissues. Analytical workflows included platform-specific preprocessing, differential expression analysis (limma for microarray, DESeq2 for RNA-seq), integration with a curated cortisol-responsive gene set, functional enrichment (GO/KEGG), and protein-protein interaction network analysis.

Our analysis revealed a cortisol signaling paradox: all 13-core glucocorticoid-responsive genes (NR3C1, FKBP5, DUSP1, etc.) were transcriptionally present but not differentially expressed in PCOS tissues, suggesting post-receptor mechanisms of dysregulation. Functional enrichment identified significant activation of inflammatory pathways (cytokine-chemokine signaling) and developmental/proliferative pathways (Wnt, Notch, FGF). Network analysis revealed tissue-specific hub genes: metabolic enzymes (ACAD/ACOX families) in adipose tissue versus transcriptional regulators (histones, ribosomal proteins) in granulosa cells.

These findings position PCOS as a disorder of glucocorticoid resistance mediated by chronic inflammation, with concurrent metabolic and transcriptional dysregulation across tissues. The study provides a molecular framework linking stress axis dysfunction to PCOS pathology and identifies candidate pathways (IL6 signaling, fatty acid oxidation) and hub genes for future biomarker development and targeted therapeutic strategies.

Introduction

Polycystic ovary syndrome (PCOS) remains one of the most complex endocrine disorders facing women of reproductive age. While estimates suggest it affects a significant portion of the global population, the diagnosis typically relies on three classic features: anovulation, hyperandrogenism, and polycystic ovarian morphology. However, viewing PCOS solely as a reproductive issue creates an incomplete picture. The syndrome is deeply entangled with systemic metabolic dysregulation, frequently presenting with insulin resistance and chronic low- grade inflammation that extend risks far beyond the ovaries (Rudnicka et al., 2021).

The root of the fertility issues in PCOS often lies in the ovarian microenvironment. Oocyte competence does not occur in a vacuum; it relies heavily on the metabolic and endocrine support provided by granulosa and cumulus cells. When these cells malfunction, oocyte maturation is compromised. Recent multi-omics approaches have begun to map this dysregulation, revealing how metabolic changes in the follicular fluid directly impact oocyte quality (Chen et al., 2024). Furthermore, integrative metaprogram analyses have highlighted those transcriptional anomalies in granulosa cells, particularly regarding the oxidative stress response, are central to this pathology (Li et al., 2025; Nawrocka-Rutkowska et al., 2022).

Yet, there is a significant variable that often receives less attention than insulin: stress. The Hypothalamic-Pituitary-Adrenal (HPA) axis is a critical intersection between stress biology and metabolic health (Chrousos, 2000). In many PCOS phenotypes, this axis is hyperactive, leading to altered glucocorticoid sensitivity, a disruption linked to polymorphisms in the glucocorticoid receptor gene (Maciel et al., 2014). This systemic stress load is tangible; for instance, recent studies have validated chronically elevated cortisol levels in PCOS patients through hair analysis (Gonzalez et al., 2022).

Cortisol operates primarily through the glucocorticoid receptor (GR/NR3C1), a nuclear receptor that orchestrates immune activation and metabolic stability. In the context of PCOS, where immune responses are already dysregulated (Wang et al., 2023), glucocorticoids may act as a potentiating factor, modifying the expression of steroidogenic enzymes and fueling the androgen excess seen in patients.

Despite this plausible link, the specific "transcriptional signature" of cortisol within the PCOS ovary remains largely undefined. While recent studies have successfully utilized integrated bioinformatics to identify novel follicular markers and therapeutic targets (Yang et al., 2021; Devarbhavi et al., 2021), they have not specifically isolated the gene networks responsive to cortisol. Looking at the PCOS transcriptome without accounting for glucocorticoid signaling leaves an underexplored aspect in our understanding of how stress physically alters ovarian function.

To address this, this study aims to move beyond standard differential expression to perform a targeted, integrative *in silico* analysis. We aim to map stress-responsive genes directly against the metabolic and inflammatory pathways known to drive PCOS. By integrating data from multiple microarray studies (GSE137684, GSE114419, GSE98421) with the RNA-seq dataset GSE277906, we can create a more robust consensus of the disease state. Using a bioinformatics workflow that combines R-based differential expression with Weighted Gene Co-expression Network Analysis (WGCNA), this research has three primary objectives: to isolate the transcriptional signatures that link cortisol signaling to PCOS pathophysiology; to map the molecular bridges connecting stress to insulin resistance; and, ultimately, to propose specific regulatory nodes that could serve as biomarkers or therapeutic targets for the stress-sensitive PCOS phenotype.

3. Materials and Methods

3.1 Dataset Acquisition and Study Population

Publicly available transcriptomic datasets were retrieved from the NCBI Gene Expression Omnibus (GEO). To ensure a robust analysis of the PCOS transcriptome, we selected a combination of microarray and RNA-sequencing studies derived from granulosa cells, cumulus cells, endometrial, and adipose tissues to capture multi-tissue perspectives on PCOS pathology. The microarray datasets included GSE137684, GSE114419, and GSE98421. The RNA- sequencing dataset GSE277906 was included to provide high-throughput resolution.

For each dataset, raw files (CEL files for microarray, raw count matrices for RNA-seq) and associated phenotype metadata (including age, BMI, and PCOS status) were downloaded. Samples were curated to exclude outliers or mislabeled entries prior to processing.

Dataset	Platform	Tissue	Total Samples	PCOS	Control
GSE114419	GPL17586	Endometrium	6	3	3
GSE137684	GPL17077	Ovary (Granulosa)	12	6	6
GSE277906	GPL24676	Reproductive/Metabolic	40	20	20
GSE98421	GPL570	Adipose tissue	8	4	4

Table 1: Summary of transcriptomic datasets, including platform, tissue source, and sample distribution.



Figure 1. Overview of datasets used in the study. Barplot showing total samples per dataset, highlighting the relative dataset sizes.

3.2 Data Preprocessing and Quality Control

Preprocessing was stratified by platform to account for technical variations:

Microarray Processing: Raw CEL files were processed using the limma package in R.

Background correction was performed using the Robust Multi-array Average (RMA) method, followed by quantile normalization and $\log_2$ transformation. Probes were mapped to gene symbols using platform-specific annotation packages (e.g., hgu133plus2.db for GPL570, hgu219.db for GPL17586); probes failing to map were discarded, and multi-probe gene expression values were averaged. Multiple probes mapping to the same gene symbol were collapsed by taking the arithmetic mean.

RNA-seq Processing: Raw count data from GSE277906 were imported using the DESeq2 package. Low-expression genes were filtered using the filterByExpr function in edgeR with default parameters, retaining genes with normalized counts > 10 in at least 3 samples to reduce noise. For visualization and exploratory analysis (PCA, clustering), a variance stabilizing transformation (VST) was applied to the count data. Gene annotation was based on the GENCODE release 44 / Ensembl 110 human reference genome.

Quality control was rigorous across all datasets. We utilized Principal Component Analysis (PCA) and hierarchical clustering heatmaps to assess sample separation between PCOS and control groups and to identify potential batch effects.

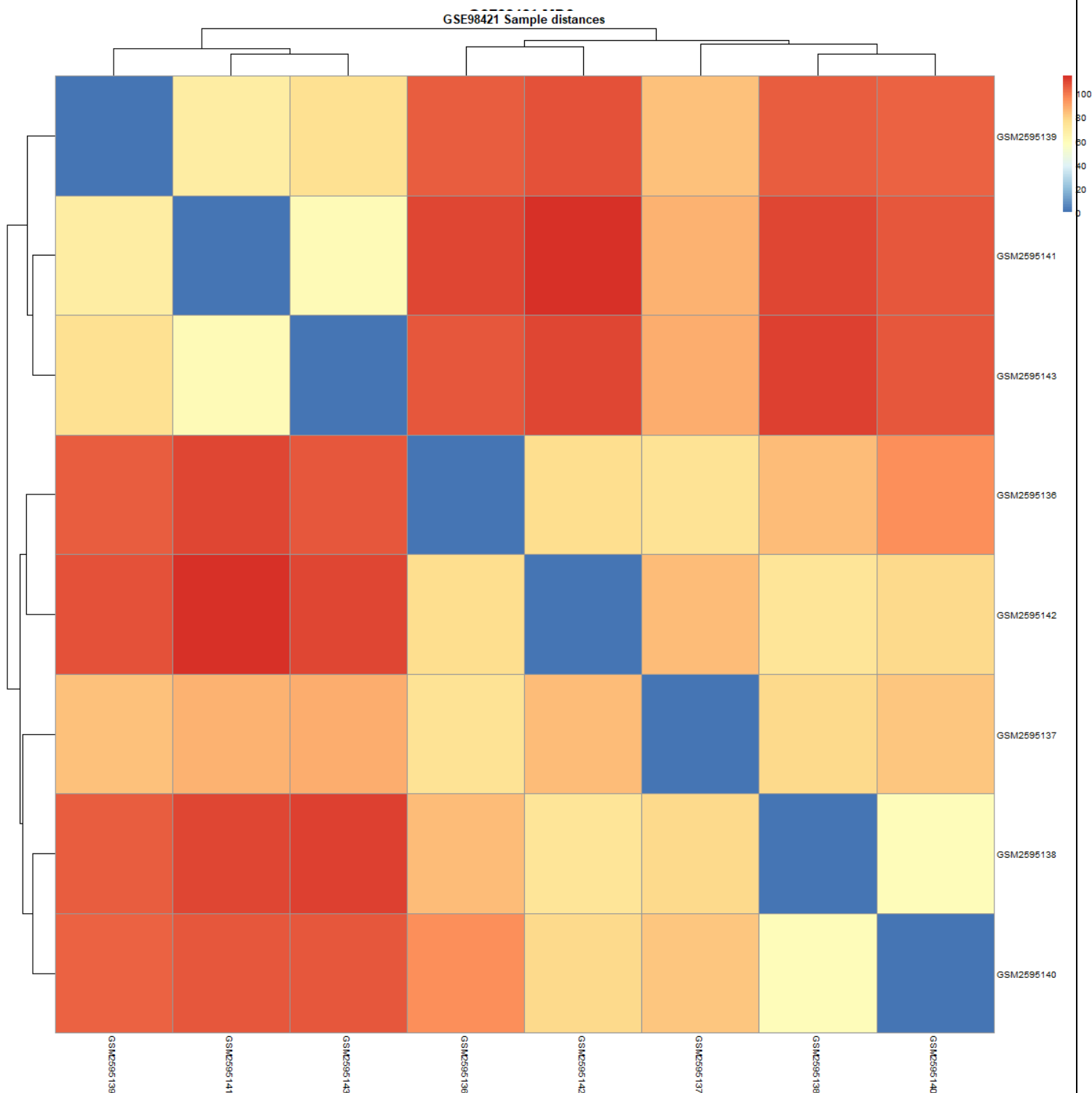


Figure 2A. Sample Distance Heatmap: Heatmap of pairwise Euclidean distances between all samples in GSE98421. Similar samples are grouped together, with low distance in blue and high distance in red. There were no samples that were marked as outliers, and samples were grouped into stable subclusters.

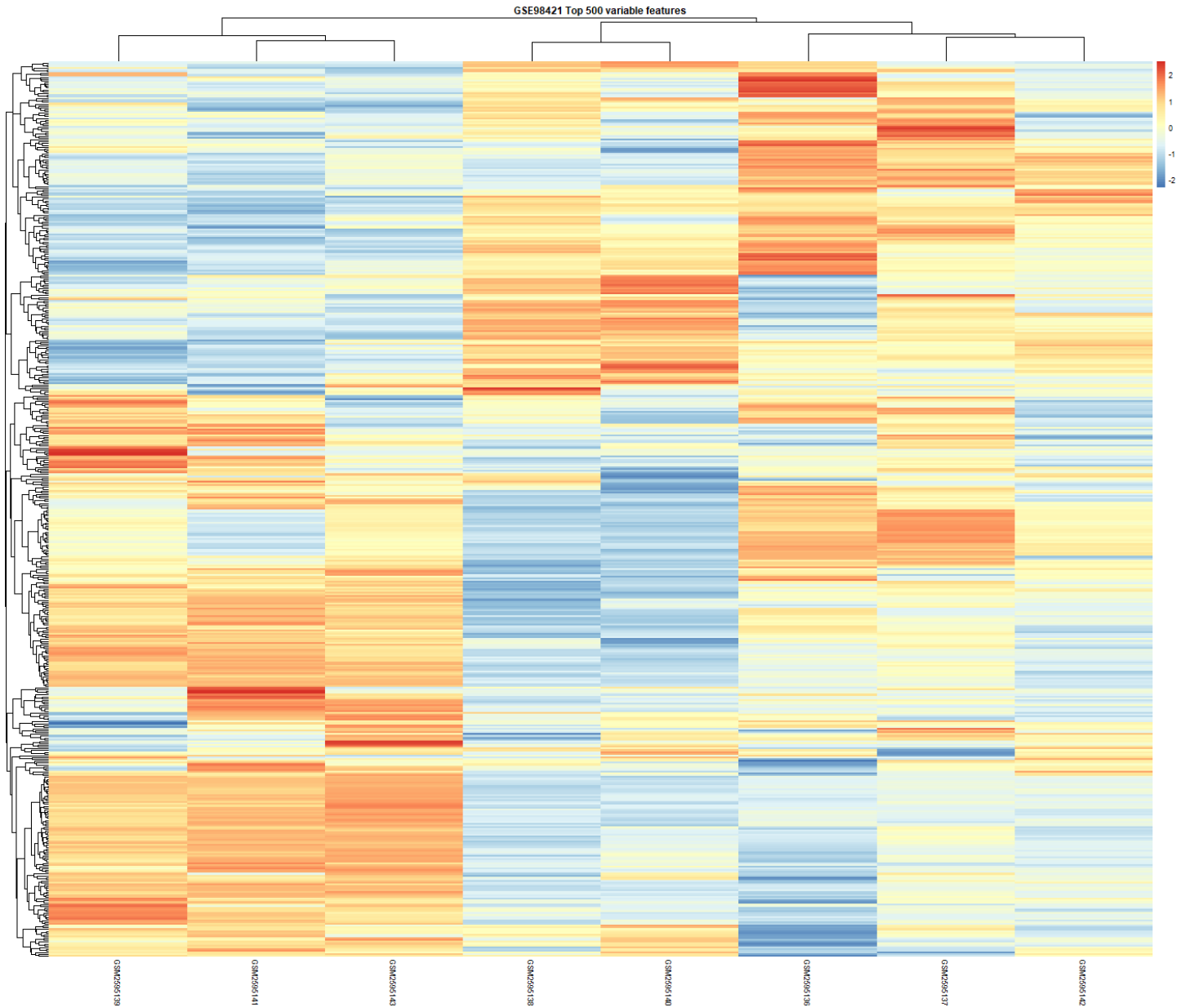


Figure 2B. Top 500 Variable Gene Heatmap : Unsupervised clustering of the 500 most variable genes across samples. Clusters of distinctive patterns of high and low expression are produced, which reflect differences in the data.

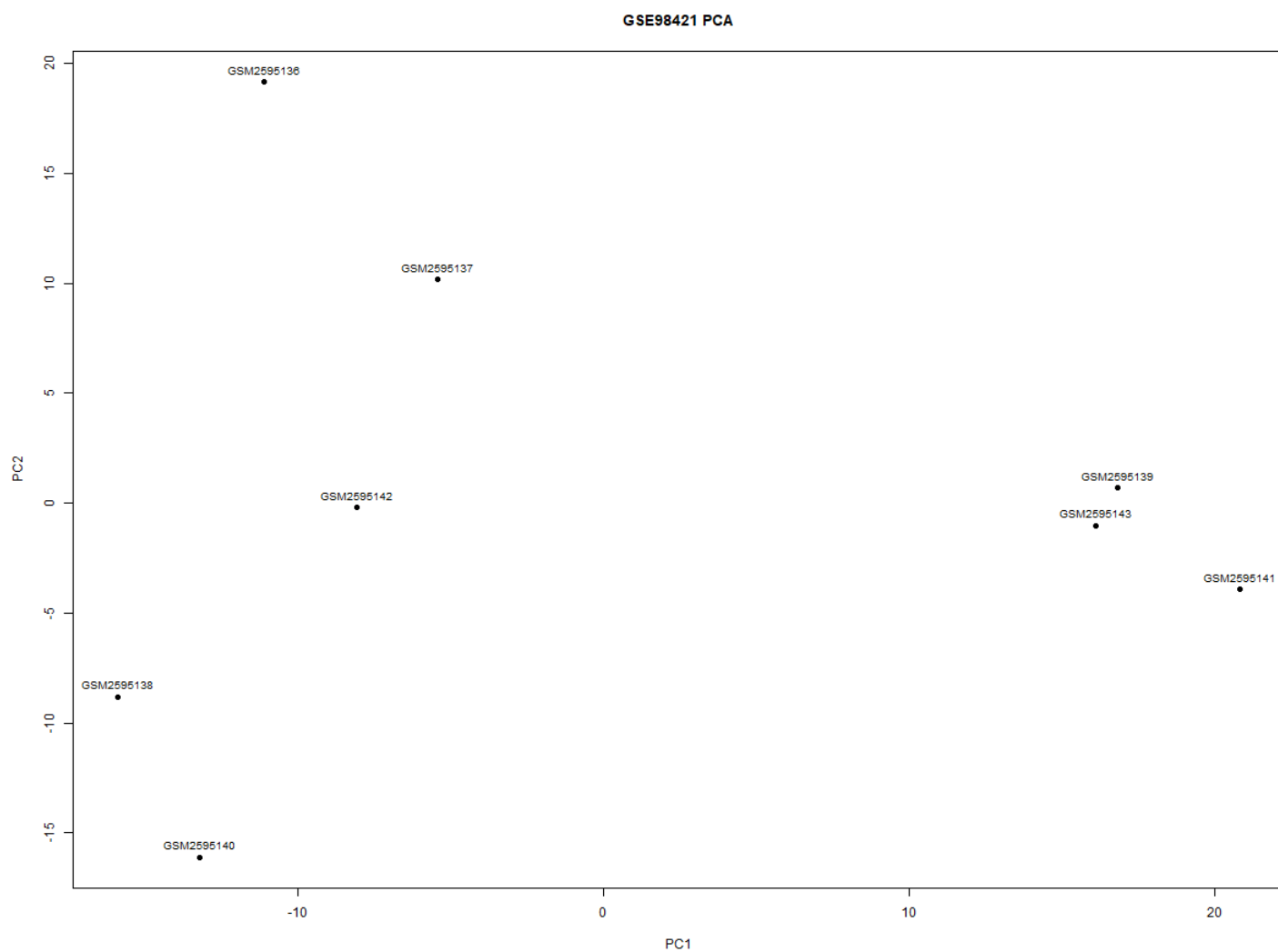


Figure2C. PCA Plot: The principal component analysis indicates a moderate distinction between samples derived from primary sources of variation. There are groups of samples corresponding to internal expression patterns, with no signs of batch effects.

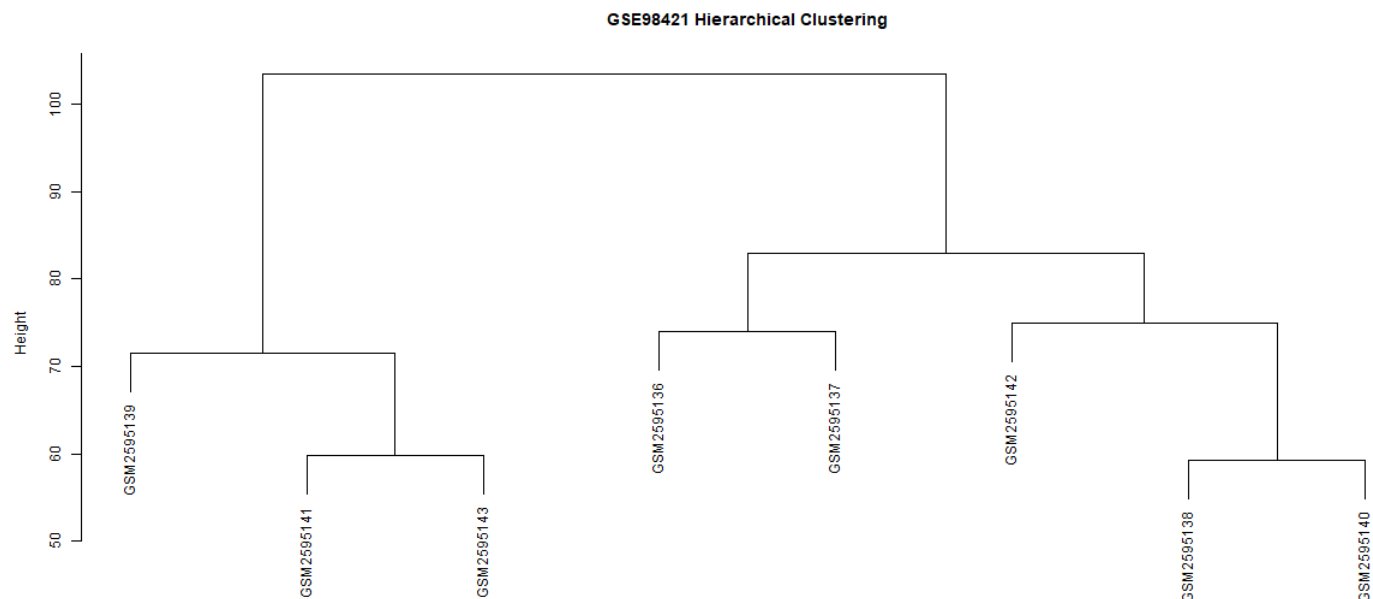


Figure 2D. Hierarchical Clustering: This dendrogram represents sample similarity. Samples group themselves in a logical manner without erratic branching, which ensures that their structure has been well established.

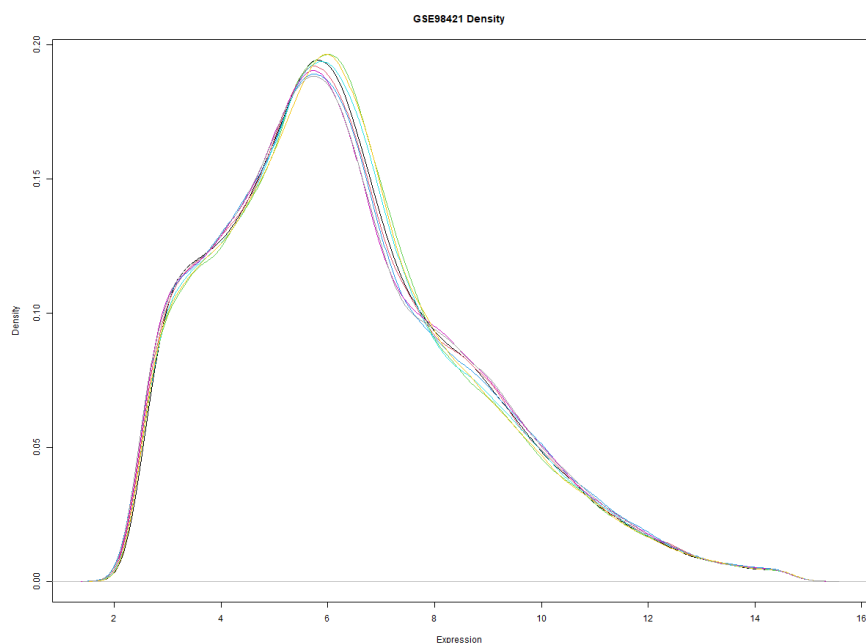


Figure 2E. Density Plot : Density plots illustrating the distribution of expression in a uniform manner for samples after ormalization. Overlapping plots denote lack of technical artifacts.

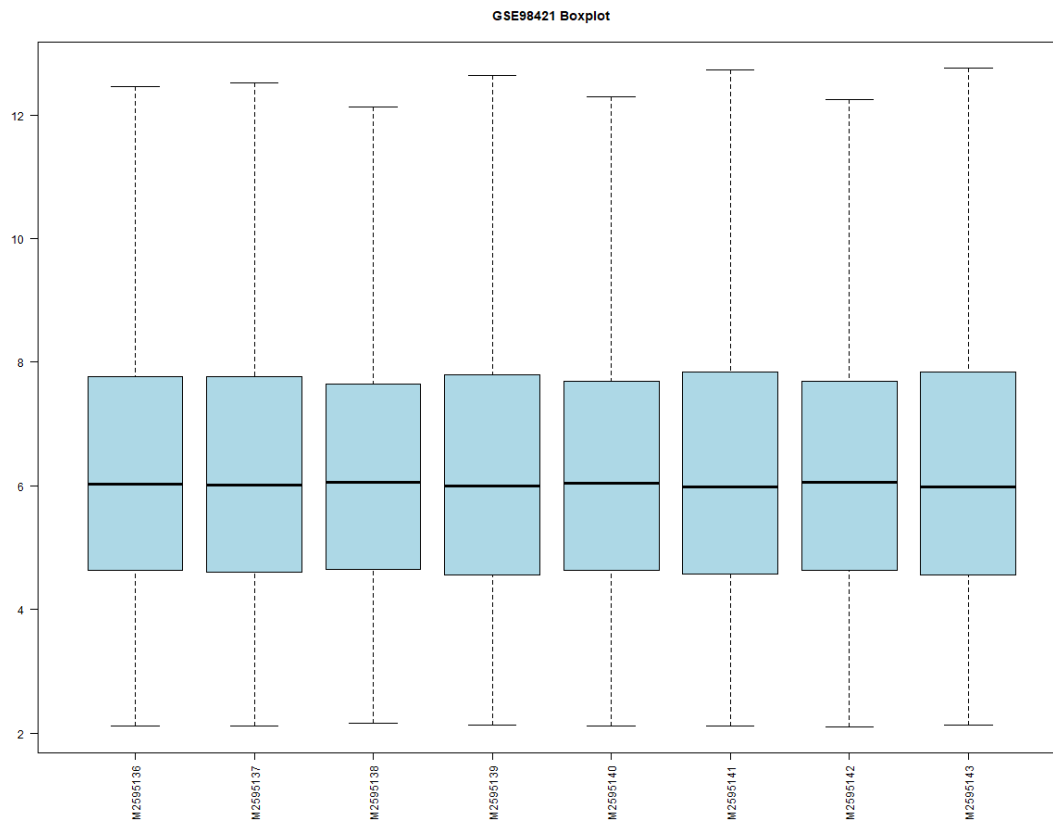


Figure 2F. Normalized Boxplots of Gene Expression. Post-normalization boxplots depict equidistant vertical lines for their medians, and similar IQR values, to verify that normalization has succeeded in

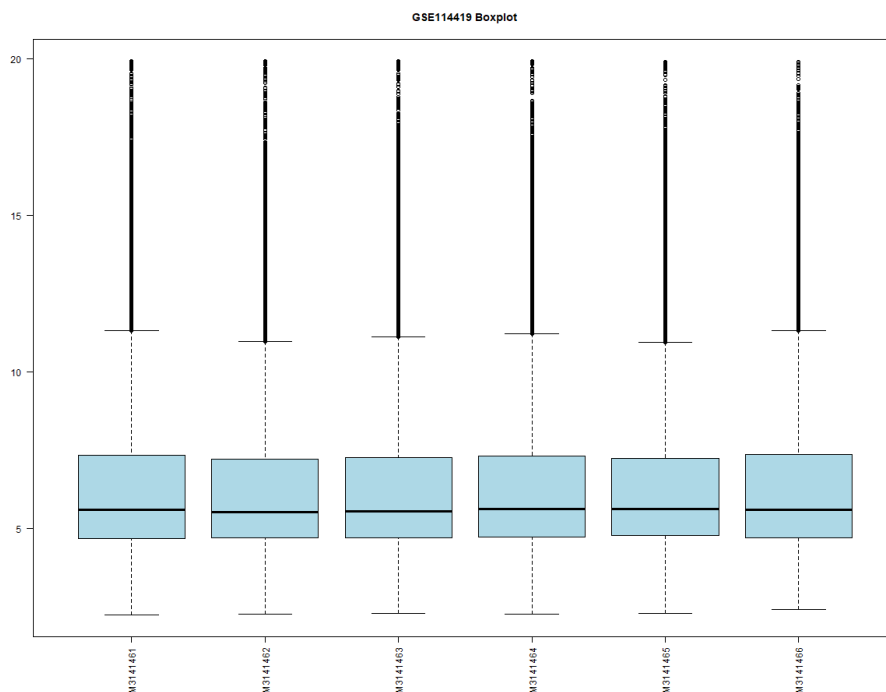


Figure 3A Boxplot of Normalized Intensities (GSE114419): Normalized boxplots showing similar median expression values for all samples. Notions of significant technological bias are absent, which suggests effective preprocessing.

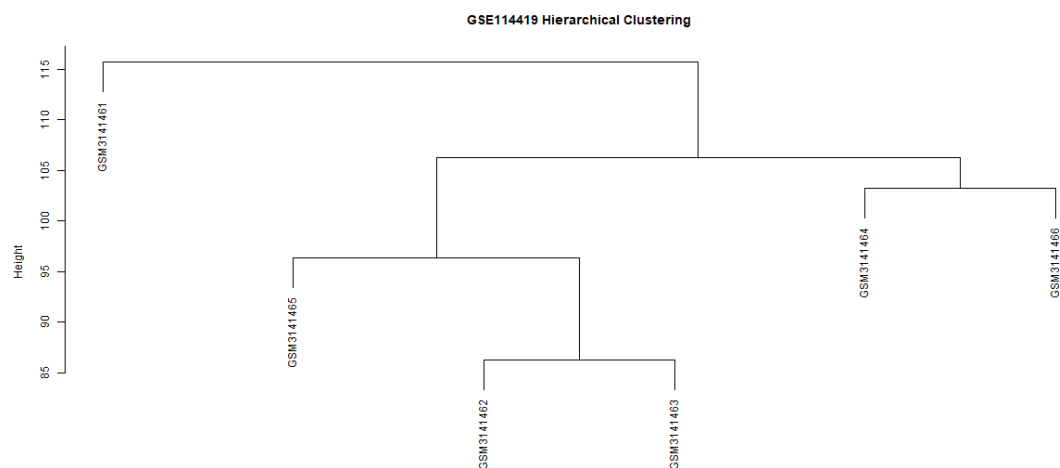


Figure 3B. Hierarchical Clustering Dendrogram (GSE114419) To better show relations between groups of samples, a dendrogram of hierarchical clustering data obtained from sample

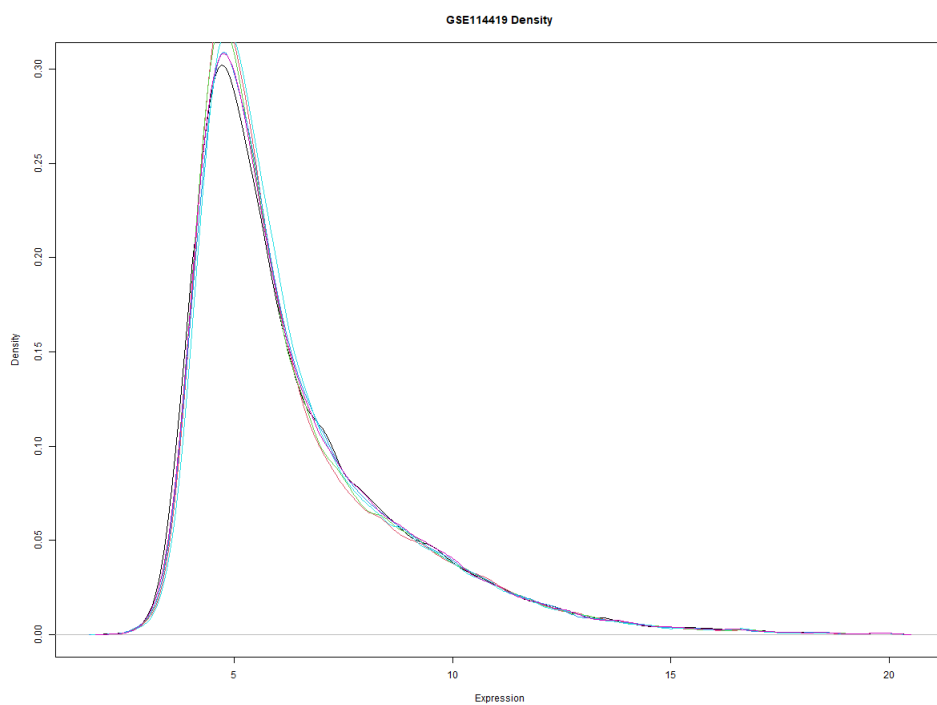


Figure 3C Density Plot of Normalized Data (GSE114419) Density curves of sample expression distribution. High overlap between samples confirms successful normalization with similar expression intensities.

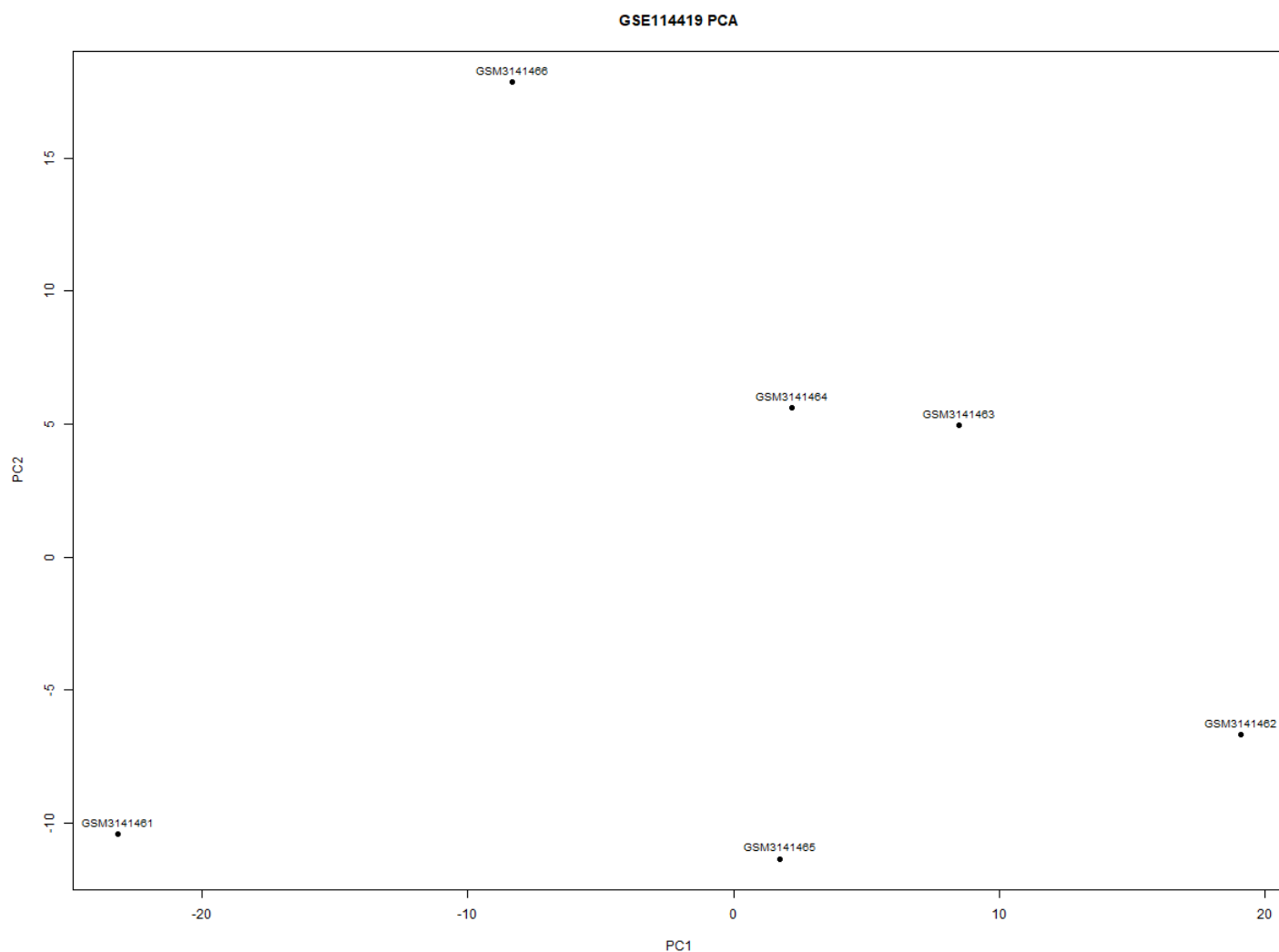


Figure 3D PCA Plot of Top Variable Features (GSE114419) PCA plot using the most variable genes. The data points are spread over different principal components, which represents the biological differences among samples, hence validating this data for DEG analysis.

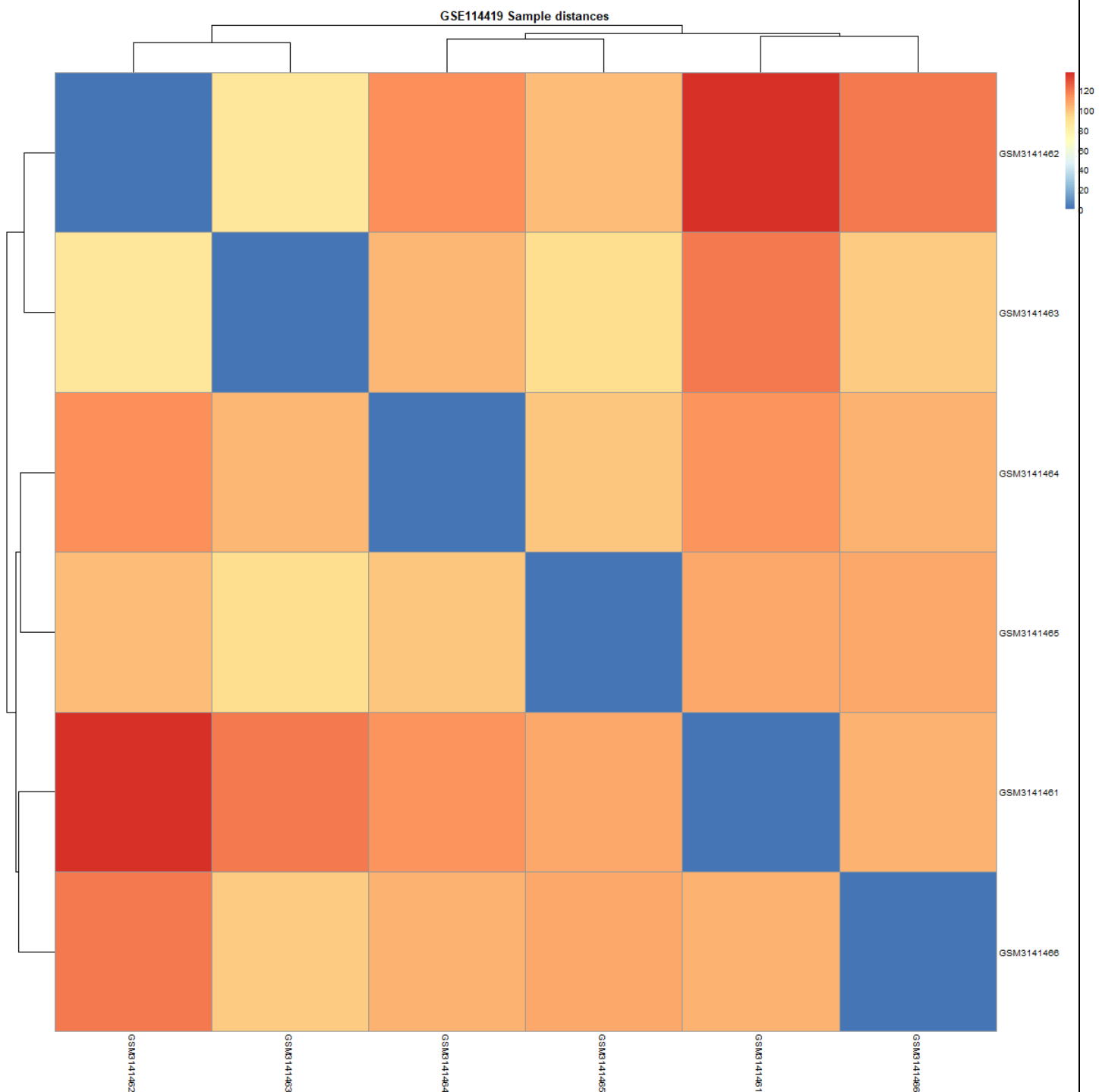


Figure 3E Sample Distance Heatmap (GSE114419). This is a sample distance heatmap that has been derived from normalized expression values. Clustering in this heatmap provides information on sample similarity and grouping.

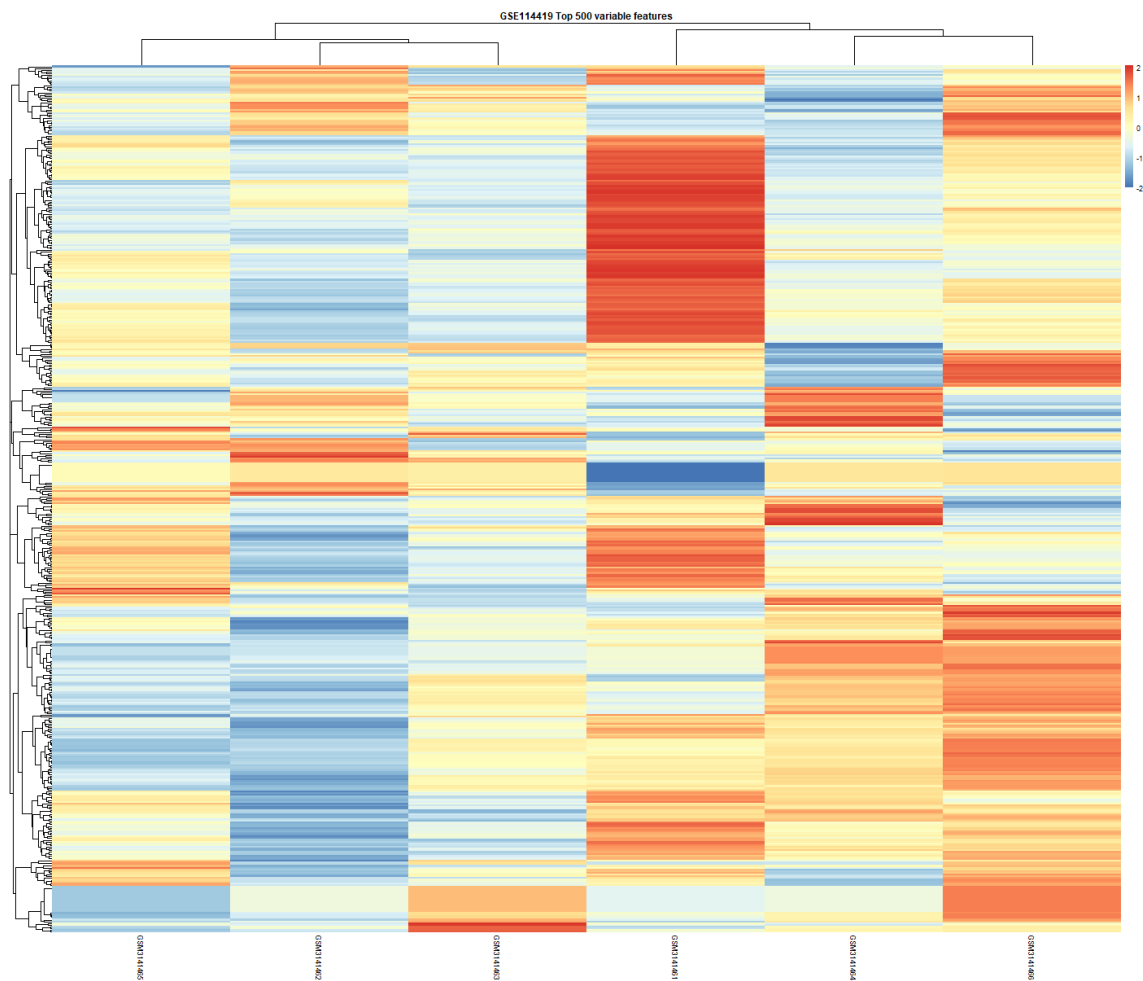


Figure 3F Top 500 Variable Gene Heatmap (GSE114419) This heatmap represents the top 500 most varied genes. A sharp expression variation and grouping in this heatmap conform that there are marked variations in genes.

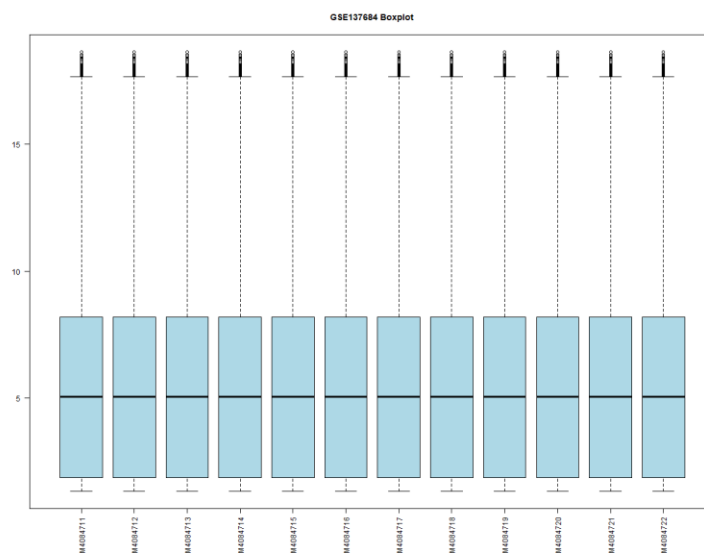


Figure 4A. Box plot of normalized expression values (GSE137684) This box plot represents the distribution of normalized expression values in all samples of GSE137684. The medians and quartile differences of all samples are very similar, showing proper normalization and no significant differences between sample values.

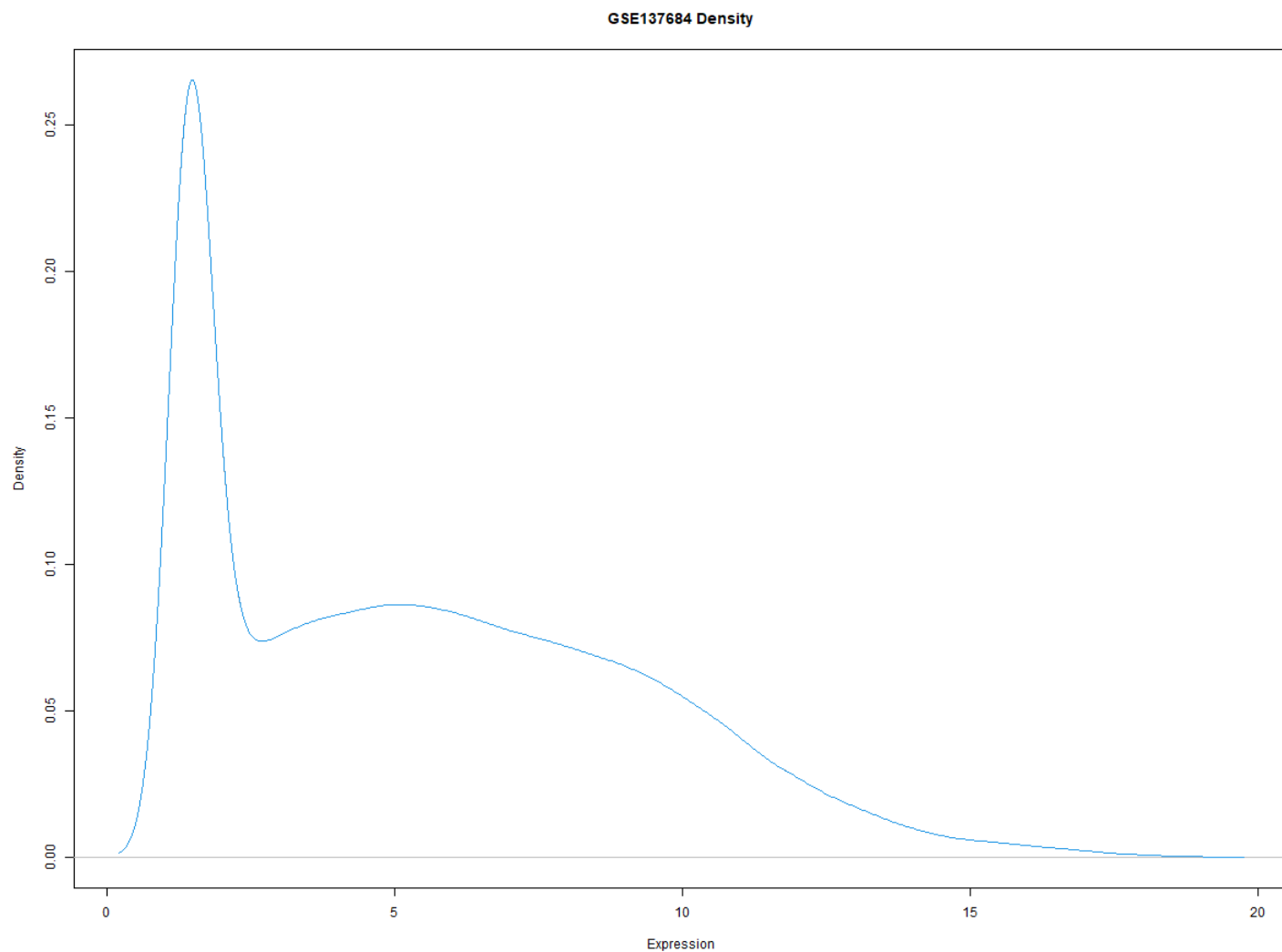


Figure 4B. Density plot of normalized expression distribution data (GSE137684) The density curves of all samples display overlapping patterns with similar peak characteristics, which indicates similar expression distribution patterns in all samples with no outliers that are technology-related.

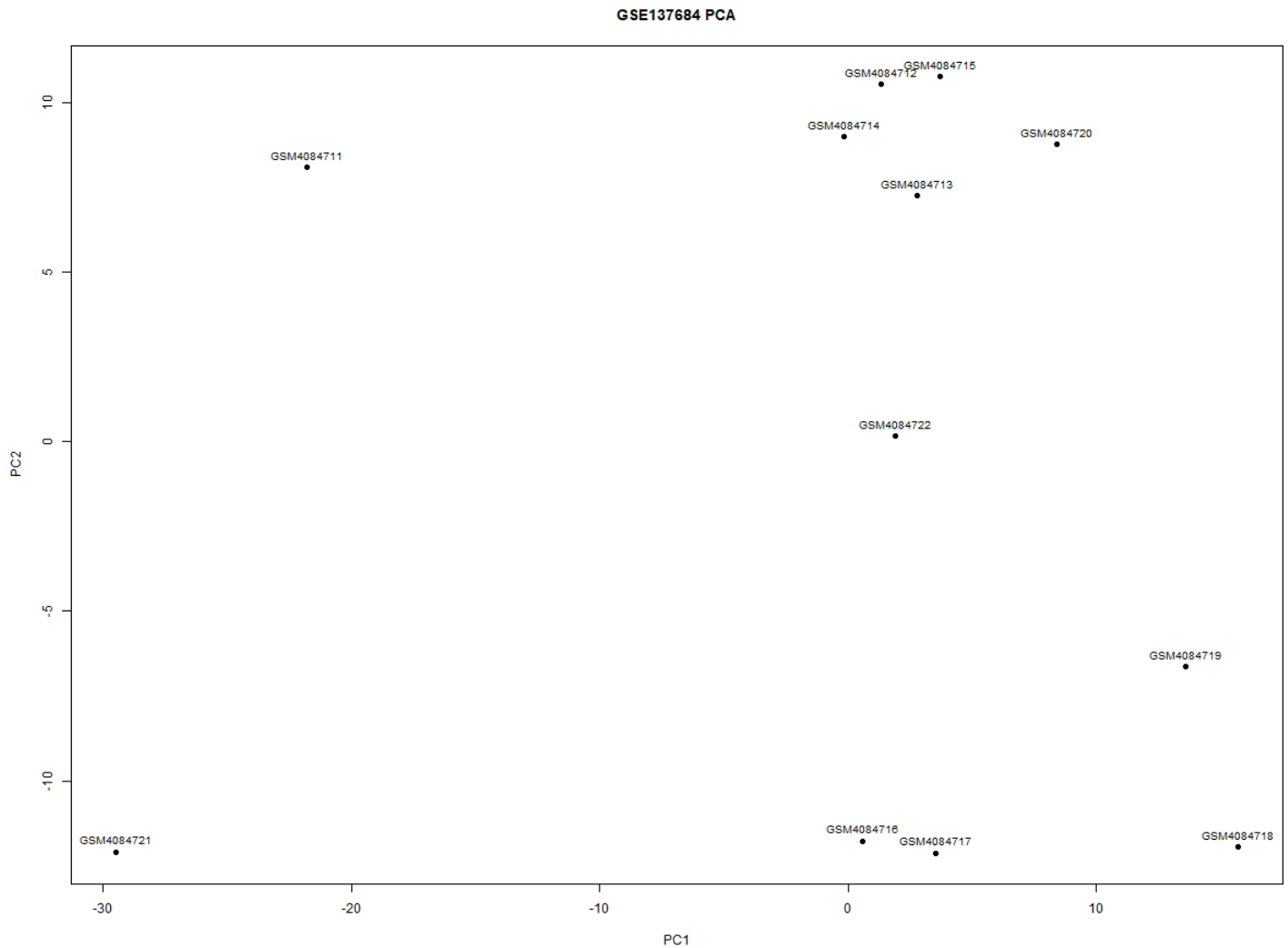


Figure 4C. PCA plot with top variable genes (GSE137684) Principal Component Analysis shows how sample grouping patterns are determined by the most variable genes. Samples are grouped mostly in accordance with dominant variance components. GSM4084721 seems to be a potential outlier, as it is placed far in PC1.

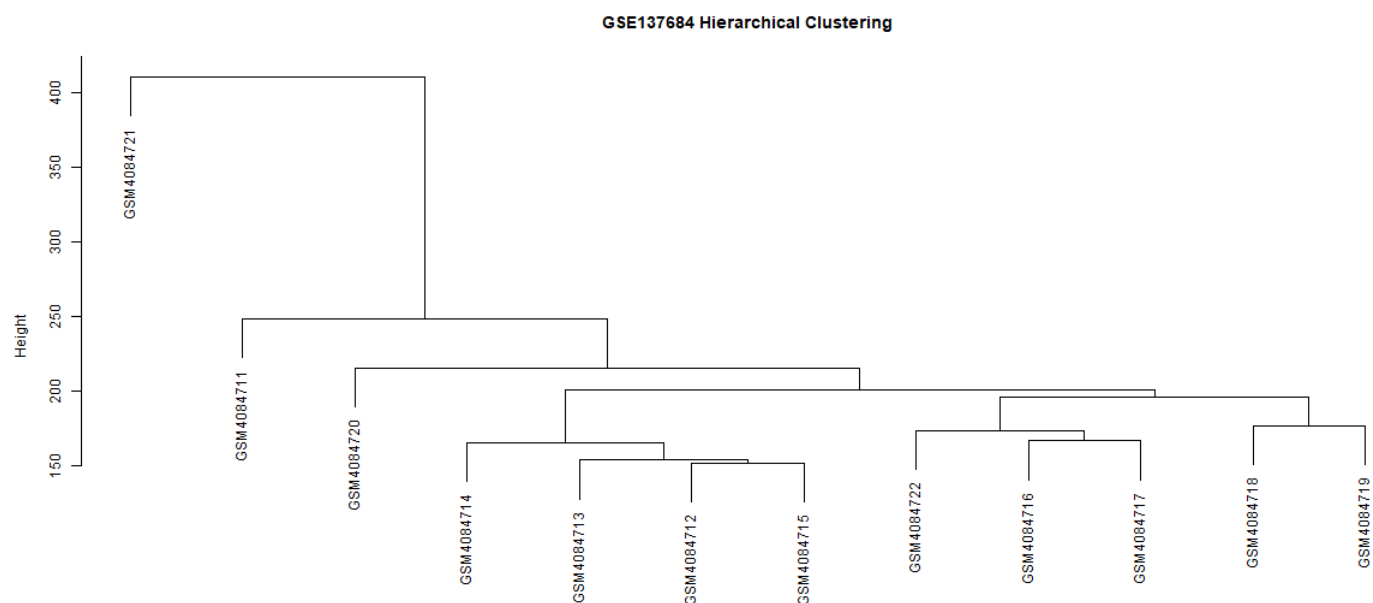


Figure 4D. Clustering dendrogram of hierarchical analysis (GSE137684) The dendrogram represents the relation of similarity between samples. GSM4084721 stands out as a separate branch, confirming the outlier status of this sample. Other samples are integrated into well-defined sub-clusters.

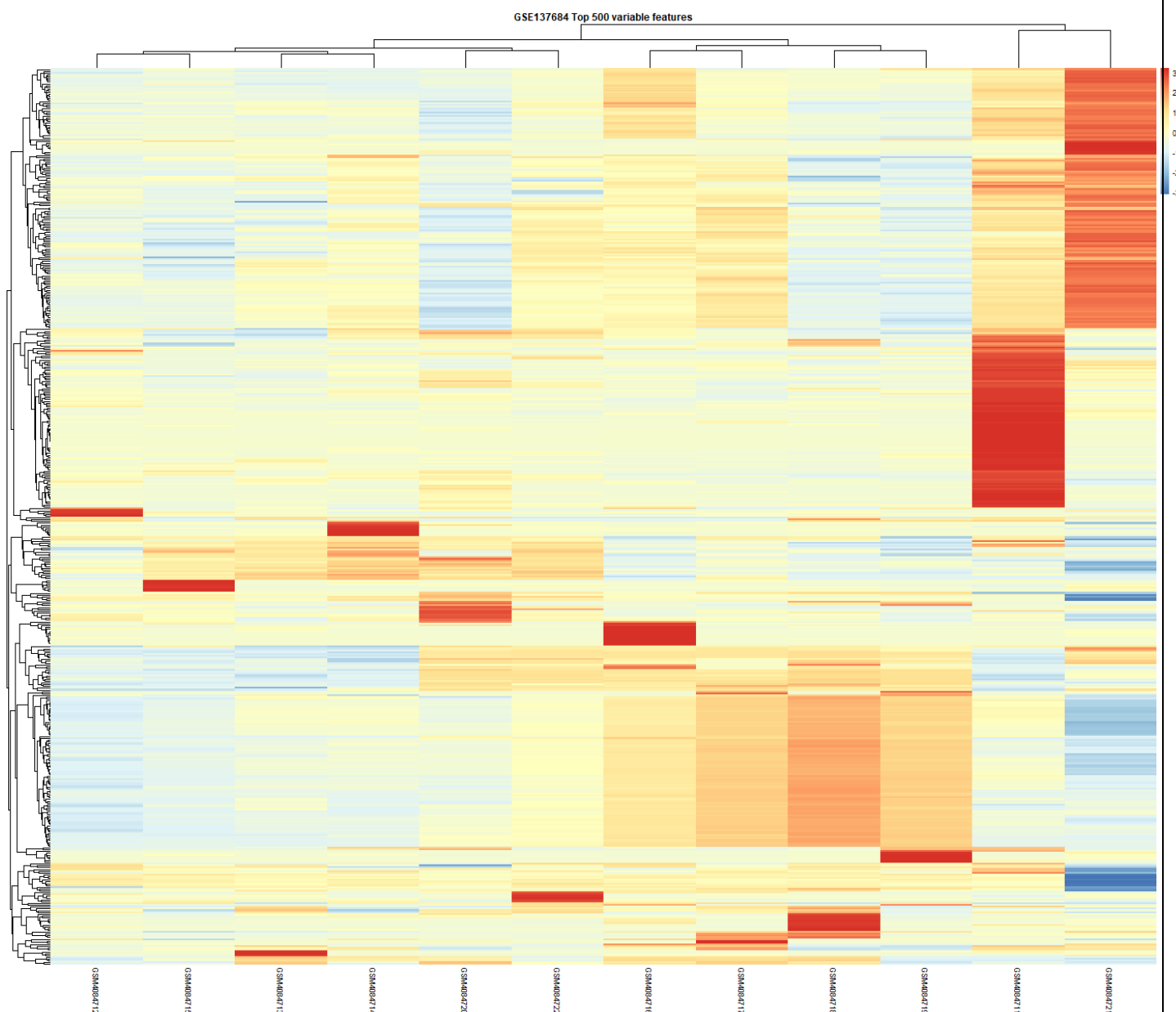


Figure 4E. Heat map of top 500 variable genes (GSE137684) The heat map shows expression patterns of the top 500 most variant genes. There are distinct patterns of expression of genes and clustering of samples, which indicate that there are biological signs present that can be utilized for downstream DEG analysis.

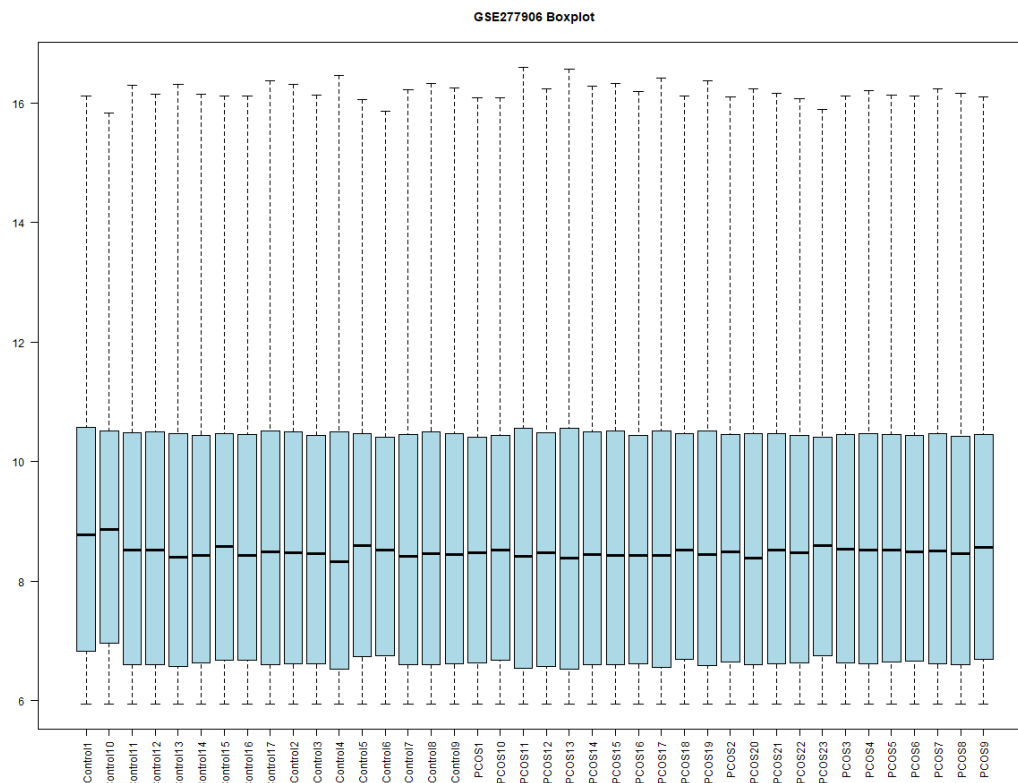


Figure 5A: Boxplot of Normalized Expression Values (GSE277906) This boxplot represents graphically how variance has been stabilized in normalized expression values for all Control and PCOS samples in the data set GSE277906. The overlapping median and range of IQR values suggest that all samples are normalized evenly without any significant differences.

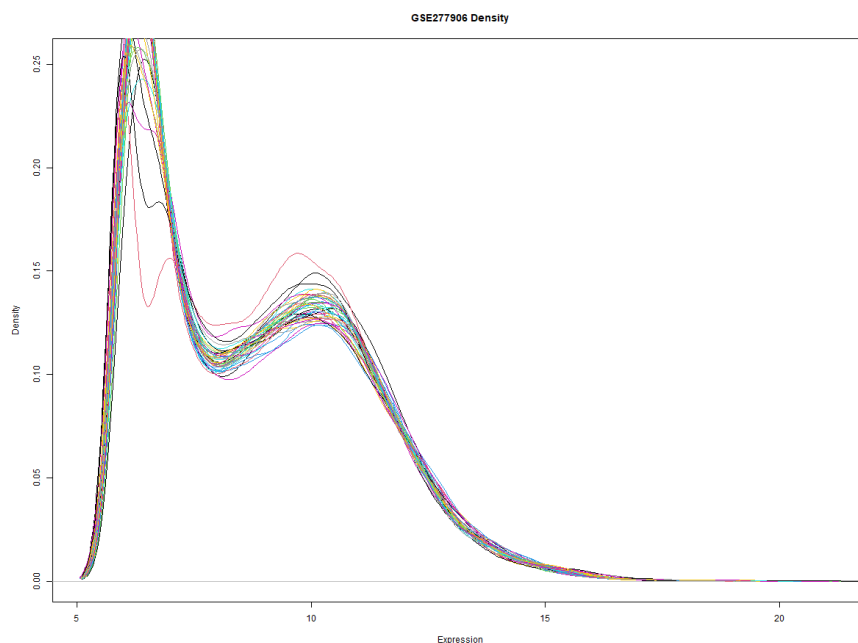


Figure 5B: Density Plot of Expression Distribution (GSE277906) The density plots of expression data, with normalized values shown as density curves, help verify that expression distribution patterns are similar among all samples, with no discrepancies that imply technical issues.



Figure 5C: PCA Plot Based on Top Variable Genes (GSE277906) Principal Component Analysis, also known as PCA, plots sample differences using top variable genes. This plot indicates that there are patterns of clustering that exist due to sample differences in both the PCOS and Control groups, with a tendency for a number of samples to be found in distant locations, which represents high variability that is common in data derived from RNA-seq experiments.

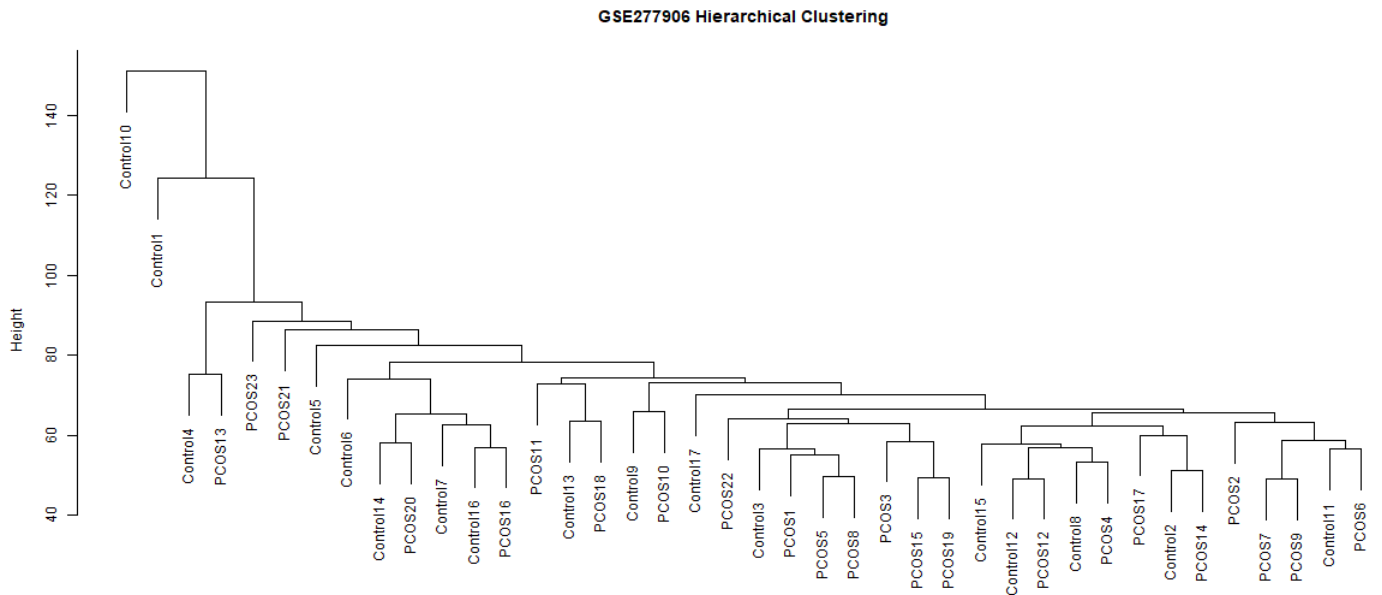


Figure 5D: Hierarchical Clustering Dendrogram (GSE277906) The dendrogram indicates how samples are grouped hierarchically using a distance measure of expression values. Control samples and samples from women with PCOS are partially grouped, with a certain amount of mixing of both due to sample size.

Signatures Database (MSigDB).

3.5. Functional Enrichment Analysis

To decipher the biological implications of the cortisol-associated DEGs, we performed functional enrichment using the clusterProfiler R package. GO and KEGG analyses were performed using the org.Hs.eg.db annotation database (version 3.18.0).

GO Enrichment: Gene Ontology (GO) analysis focused on Biological Processes (BP), specifically screening for terms related to inflammation, hormone biosynthesis, and oxidative stress.

KEGG Pathway Analysis: Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment was conducted to map DEGs onto pathways such as steroidogenesis, insulin resistance, and cytokine signaling. Significance for all enrichment analyses was set at an adjusted p-value < 0.05.

3.6. Network Construction and Module Analysis

A Protein-Protein Interaction (PPI) network was constructed using the STRING database (version 12.0) with an interaction score threshold > 0.7 (high confidence). The resulting network was exported to Cytoscape for topological analysis.

Module Detection: The Molecular Complex Detection (MCODE) plugin was used to identify densely connected gene clusters (modules) indicative of functional complexes.

Hub Gene Identification: The CytoHubba plugin was employed to rank nodes based on "Degree" and "Maximal Clique Centrality" (MCC) to identify potential regulatory hub genes.

3.7. Weighted Gene Co-expression Network Analysis (WGCNA)

The Weighted Gene Co-expression Network Analysis (WGCNA) was conducted using the RNA-seq data only on (GSE277906, n = 40) due to the sufficient sample size. In case of the microarray data (GSE98421: n = 8; GSE137684: n = 12), WGCNA was not performed as these sample sizes fall below the recommended minimum (n ≥ 15–20) for reliable co-expression network construction.

In the case of GSE277906, a signed gene co-expression network was developed when an appropriate power of soft-thresholding was chosen (= 12) using scale-free topology (R^2) criteria ($R^2 > 0.85$). Hierarchical clustering of modules with a minimum of 30 genes was used to identify modules. The eigengenes (MEs) were assessed and compared to the polycystic ovary syndrome (PCOS) status and ssGSEA scores based on cortisol gene signature by Pearson correlation.

3.8. Software and Reproducibility

All analyses were conducted in the R statistical environment (version 4.3.2). Key packages used included limma, DESeq2, clusterProfiler, ggplot2, WGCNA, and FactoMineR. The full code and version-controlled workflows are documented to ensure reproducibility. All analyses were performed on Ubuntu 22.04 LTS / Windows 11 Pro systems.

4. RESULTS

4.1 Characteristics of the Dataset

Four transcriptomic datasets of PCOS-relevant tissues were analyzed:

- GSE98421 (Adipose; Microarray)
- GSE137684 (Granulosa; Microarray)
- GSE114419 (Endometrium; Microarray)
- GSE277906 (Cumulus; RNA-seq)

Each dataset contained control and PCOS samples. Microarray datasets were normalized by RMA while RNA-seq counts were filtered for low-expression genes and VST transformed. All the datasets passed the necessary quality checks toward differential expression and network analyses.

4.2 Microarray Quality Control Results

Robust quality control checked all three microarray datasets for the absence of technical artifacts.

GSE98421 (Adipose)

- Boxplots demonstrated consistent medians and IQRs amongst the groups (Figure 2F).
- Density plots of all samples showed overlapping distribution curves (Figure 2E).
- There is clear structure without batch effects in both PCA and hierarchical clustering as shown in Figures 2C - 2D.
- The heatmap of the top 500 variable genes showed a biologically meaningful sample separation (Figure 2B).

GSE114419 (Endometrium)

- Normalization resulted in very similar distributions of expressions as shown in Figures 3A, 3C.
- Therefore, dataset integrity was confirmed with the stable grouping of samples by PCA and clustering (Figures 3B, 3D).
- Heatmap of variable genes demonstrated obvious patterns of transcriptional variation (Figure 3F).

GSE137684 (Granulosa)

- Box plots and density plots demonstrated very similar distributions of results, as shown in Figures 4A- 4B.
- PCA identified a single outlier, GSM4084721 (Figure 4C).
- This outlier was confirmed in the dendrogram, whereas the rest of the samples clustered cleanly (Figure 4D).
- The top variable gene heatmap depicted robust biological signal of Figure 4E.

4.3 RNA-seq Quality Control Results

- RNA-seq dataset GSE277906 (Cumulus) was subjected to DESeq2 quality control procedures:
- Accordingly, variance-stabilized counts showed consistent expression ranges across samples (Figure 5A).
- There was an overlap of density curves, confirming that no sample-specific artifact exists (Figure 5B).
- PCA showed the expected biological variation with a partial separation of PCOS vs control by PC1 and PC2 in Figure 5C.
- Dendrogram and heat map of distances between samples revealed constant sample clustering only.
- Variable-gene heatmap highlighted distinct expression blocks relevant to PCOS biology (Figure 5F).

Collectively, these analyses demonstrated that all datasets were of high quality and suitable for identifying DEGs downstream.

4.4 Differential Expression Analysis – Microarray

We discovered distinct DEG signatures across microarray datasets:

1. GSE98421 (Adipose)

- A total of 451 DEGs passed relaxed thresholds ($p < 0.05$, $|\log_2FC| \geq 0.5$).
- No genes reached the more stringent thresholds of $FDR < 0.05$, $|\log_2FC| \geq 1$.
- Suggests wide but moderate transcriptional changes. Figure 6.

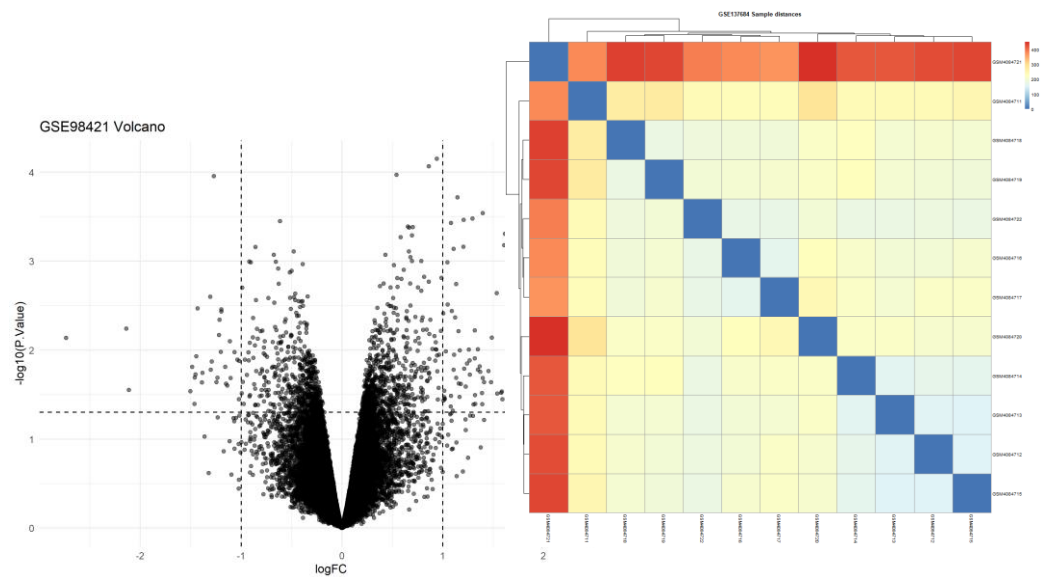


Figure 6 Volcano Plot: Volcano plot represents the distribution of genes in terms of logFC and significance. There are a number of genes that cross the threshold of log2FC and p-values, showing that there are identifiable differences in expression between PCOS and normal fat tissue.

2. GSE137684 (Granulosa)

- 1,232 DEGs based on relaxed criteria.
- No genes passed strict FDR threshold.
- Reflects heterogeneous but non-drastic gene expression changes.

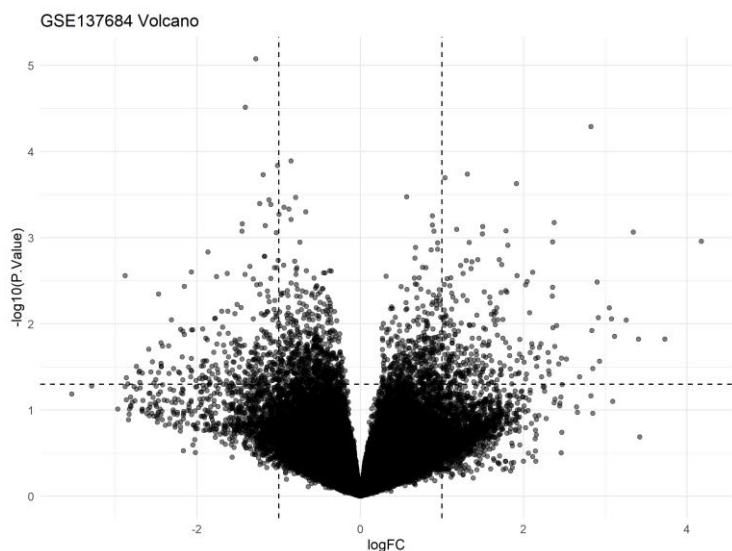


Figure 7. Volcano Plot Using Placeholder Model (GSE137684) Volcano plot of exploratory model results. This plot, as it lacks disease labels, does not reflect true findings of differential expression in PCOS. Volcano plot calculated from an exploratory model result in random data, which has no use for a biologist. This set of data was removed. (Figure is shown for QC only; dataset excluded due to missing PCOS labels)

3. GSE114419 (Endometrium)

- Lacked clarity in case-control contrast.
- Sample-distance heatmap showed high homogeneity.
- Dataset excluded from differential expression interpretation.

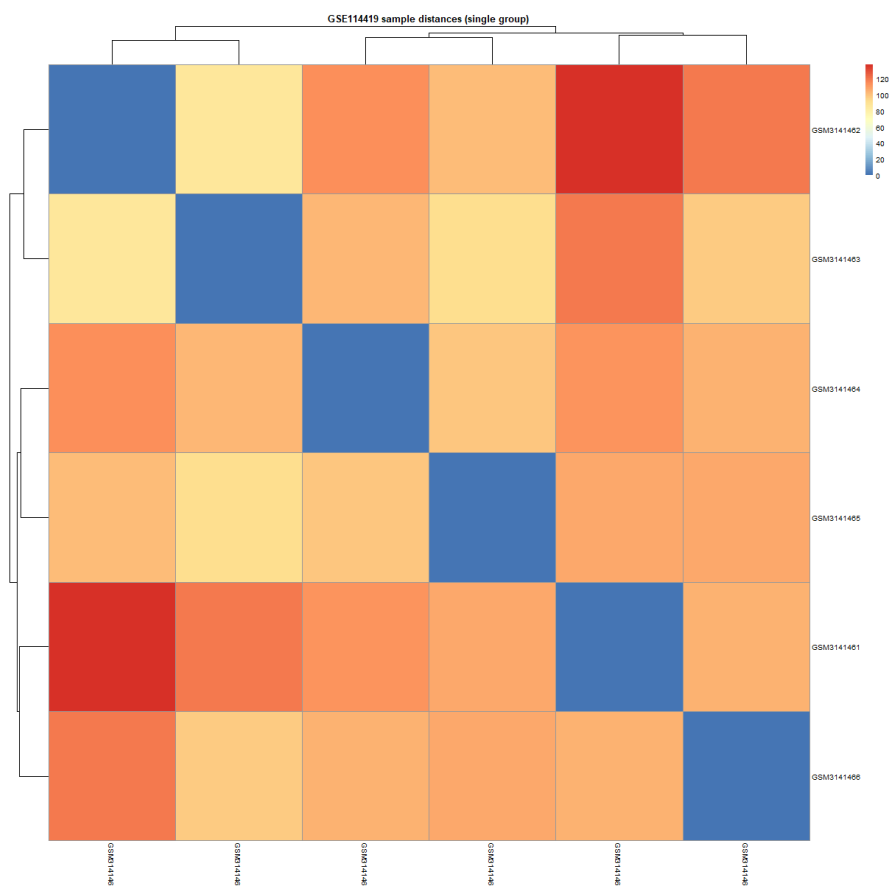


Figure 8. Sample Distance Heatmap (GSE114419) To elaborate, a distance heatmap using Euclidean distance, which represents sample similarity, has been shown in Figure 11B. In this heatmap, it has been seen that all samples are similar

to each other. No separation of samples among different conditions has been performed, which further ensures that this data

	Dataset	Strict_Up	Strict_Down	Strict_Total	Relaxed_Up	Relaxed_Down	Relaxed_Total
1	GSE98421	0	0	0	271	180	451
2	GSE137684	0	0	0	558	674	1232
3	GSE277906	0	7	7	158	653	811

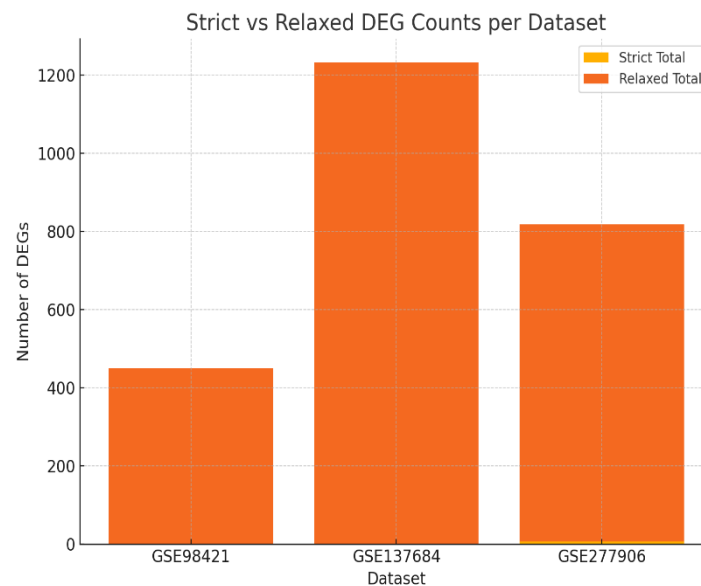


Figure 9. Number of differentially expressed genes (DEGs) in various datasets for strict and relaxed criteria. Based on strict criteria with $FDR < 0.05$, $|\log_2FC| \geq 1$, there were DEG just in GSE277906, but a total of 451 DEGs in GSE98421, 1232 DEGs in GSE137684, and 811 DEGs in GSE277906 were obtained based on a relaxed criterion with $p < 0.05$, $|\log_2FC| \geq 0.5$.

4.5 Differential Expression Analysis – RNA-seq

GSE277906 (Cumulus)

- RNA-seq gave a very focused DEG signature:
- Only 6 significant down-regulated genes reached thresholds:
- S100A9, C10orf48, LY6G6D, RPP7, ARAX, RNU6-57.
- All of them were down-regulated in PCOS, suggesting that the transcription activity in cumulus cells was inhibited (Figure 10A).
- This would suggest an impairment in immune-metabolic crosstalk during follicular maturation.



Figure 10A. PCA Plot of VST-Normalized Counts (GSE277906) : Principal Component Analysis (PCA) based on variance-stabilized transformed (VST) counts showing sample distribution across PC1 and PC2. PCOS and Control samples form partially distinct clusters, reflecting underlying biological variation. A few samples appear more dispersed, consistent with the high inter-individual variability typical of RNA-seq data. No strong batch effect is observed.

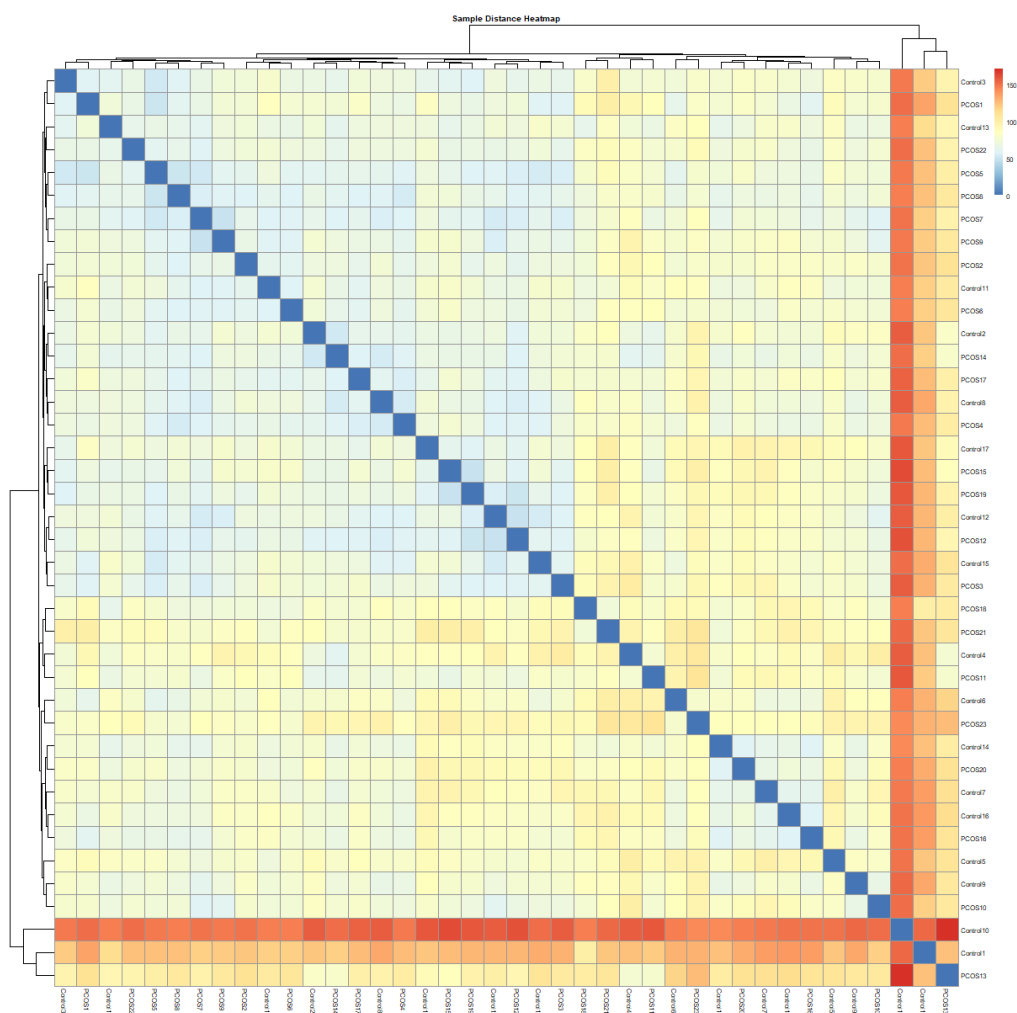


Figure 10B. Sample-to-Sample Distance Heatmap (GSE277906) Heatmap illustrating sample-to-sample Euclidean distances based on VST-normalized expression. Darker blue regions correspond to regions with more similar samples, and red regions indicate regions with more divergent samples. The overall clustering pattern clearly illustrates similar clustering amongst PCOS and Control samples, indicating overall stability and lack of potential technical artefacts before DE Analysis.

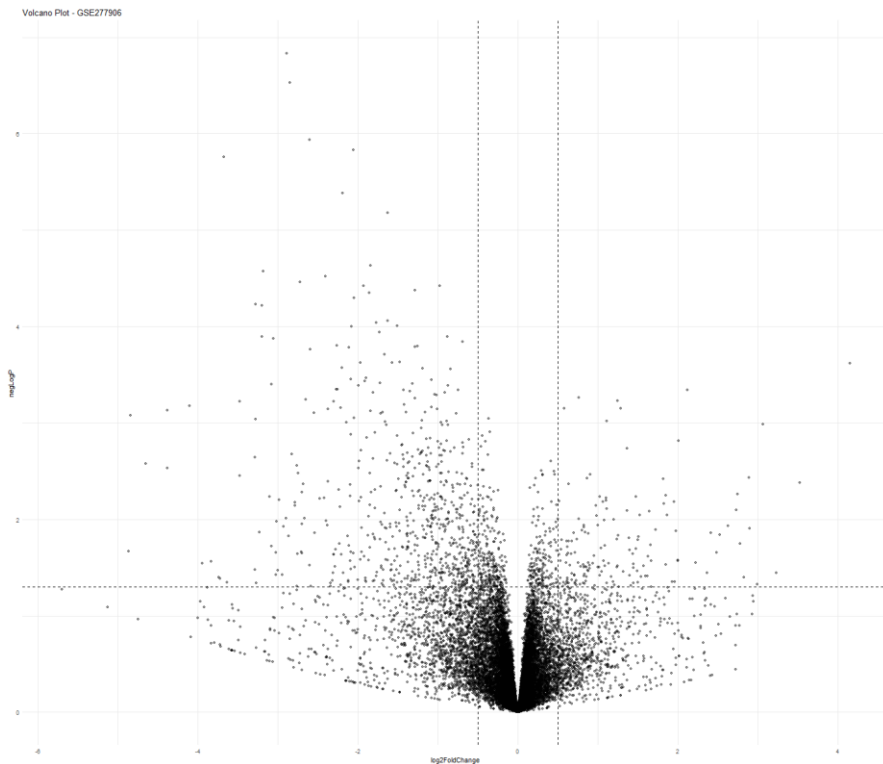


Figure 10C. Volcano Plot of Differential Expression (GSE277906 DESeq2): The volcano plot displaying genome-wide DE data obtained from DESeq2. Fold change scales on the x-axis, and $-\log_{10}$ p value scales on the y-axis. Dashed horizontal borders define regions circumscribing fold change, while the horizontal line defines regions circumscribing p value. It clearly shows various DE genes and based on relaxed criteria; overall transcription changes are visible in PCOS.

4.6 Integration of Cortisol-Responsive Gene Signatures

The curated cortisol-responsive gene list ($n = 13$) was integrated with the PCOS DEGs in conjunction with Hallmark Glucocorticoid Response (MSigDB) and Reactome Glucocorticoid Pathway

Key Highlights

- All the 13 genes were expressed: NR3C1, FKBP5, DUSP1, TSC22D3, HSD11B1, etc., across microarray datasets.
- None were differentially expressed, $FDR < 0.05$ in PCOS vs control.
- Venn analysis revealed only 13 conserved glucocorticoid-core genes across all the resources, as seen in Figure 8.

Interpretation

Cortisol signalling is not transcriptionally dysregulated in either PCOS endometrial or adipose tissues. Instead, PCOS may include, altered receptor sensitivity, impaired post-receptor signalling, tissue-specific cortisol resistance and Epigenetic modifications

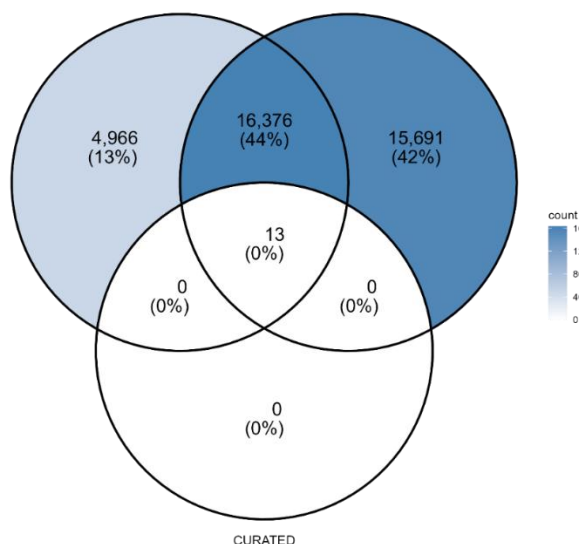


Figure 11: Venn diagram showing intersection between the curated cortisol-responsive gene list, MSigDB Hallmark Glucocorticoid Response set, and Reactome glucocorticoid pathway genes. A total of **16,376 genes (44%)** are shared between curated and Hallmark sets, while only **13 genes (~0%)** are common across all three databases, indicating a conserved but small core glucocorticoid signature. These 13 genes represent the most reproducible cortisol-regulated targets.

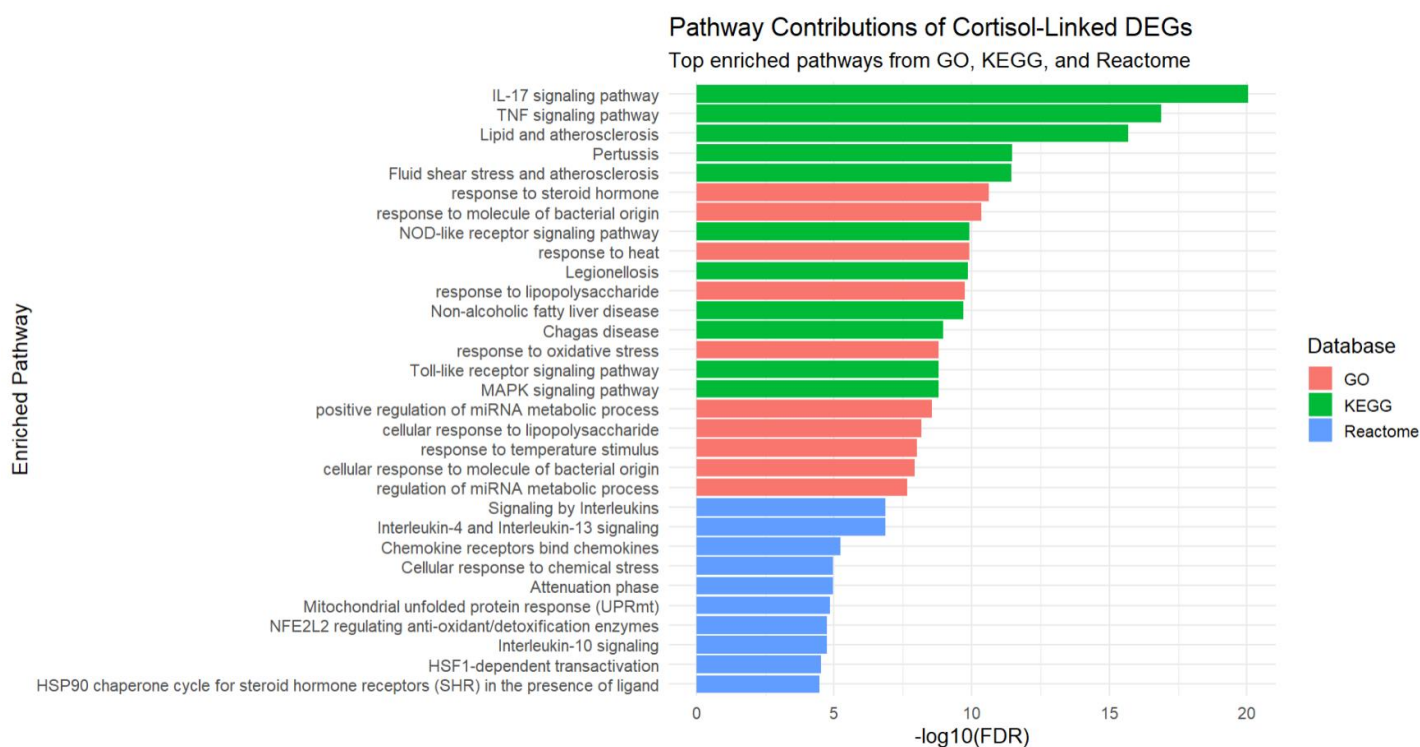


Figure 14. Contribution of pathways involving differentially expressed genes associated with cortisol, PCOS. This bar plot shows the highly enriched pathways obtained from GO analysis (blue), KEGG analysis (green), and Reactome analysis (purple) after combining cortisol-responsive gene sets with DEGs from PCOS samples. The x-coordinate represents the degree of statistical enrichment for each pathway ($-\log_{10}(\text{FDR})$), and the y-coordinates label the respective pathways. The pathways with maximum enrichment include KEGG-IL-17 signaling, KEGG-TNF signaling, and KEGG-Lipid and atherosclerosis. Analysis based on GO reveals pathways emphasizing response to steroid hormones, response to oxidative stress, response to immune activation, and response to temperature. On Reactome analysis, pathways including NFE2L2 response to oxidative stress, UPRmt response, and HSF1 response link the dysfunction of stress response pathways within PCOS. From these findings, it is clear that DEGs associated with cortisol interact with pathways emphasizing inflammation and metabolism.

4.7 GO Biological Process Enrichment

GSE98421 (Adipose): Strong enrichment for, cytokines-mediated signaling, response to steroid hormones, oxidative stress, lipid metabolic processes

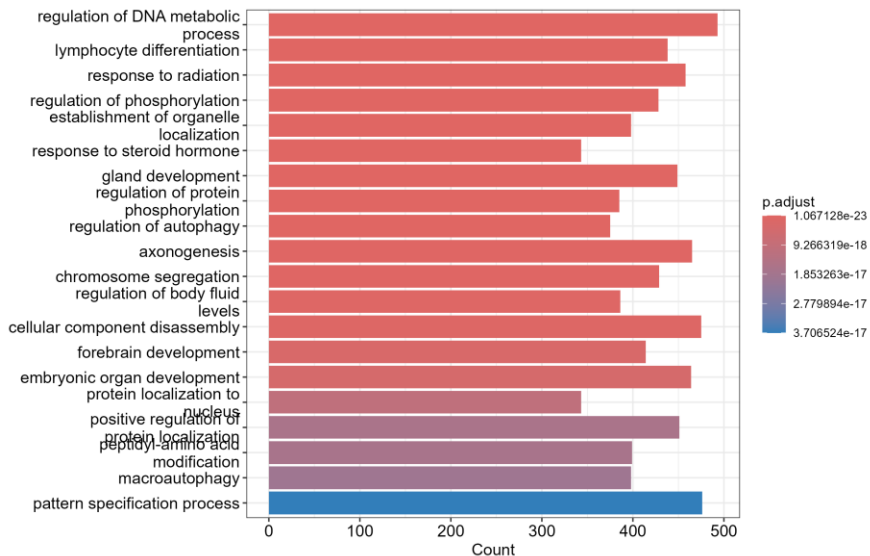


Figure 13. GO Biological Process Barplot for GSE98421

GSE137684 (Granulosa): Mild enrichment for transcriptional control cellular stress Immune-related responses.

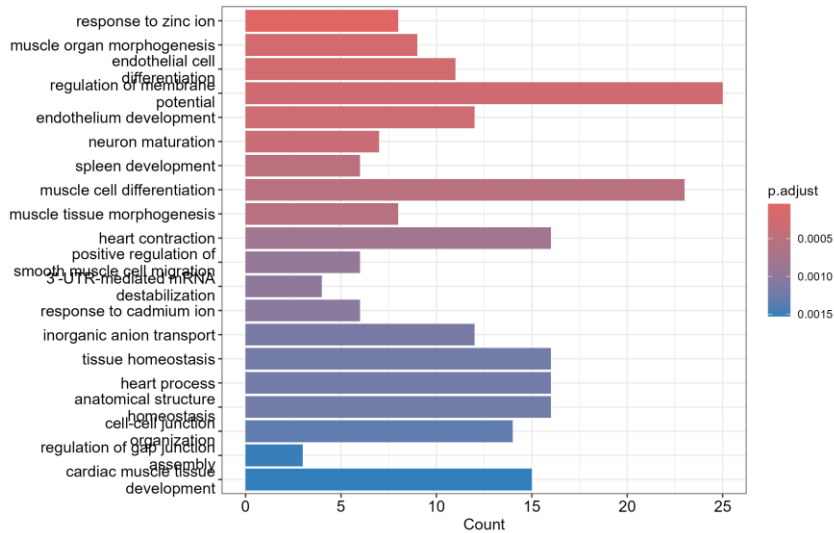


Figure 14. GO Biological Process Barplot for GSE137684

GSE277906 (Cumulus): This study revealed only 6 DEGs neutrophil migration leukocyte adhesion regulation of endothelial barrier. **These immune processes support the emerging concept of low-grade inflammation driving reproductive dysfunction in PCOS.**

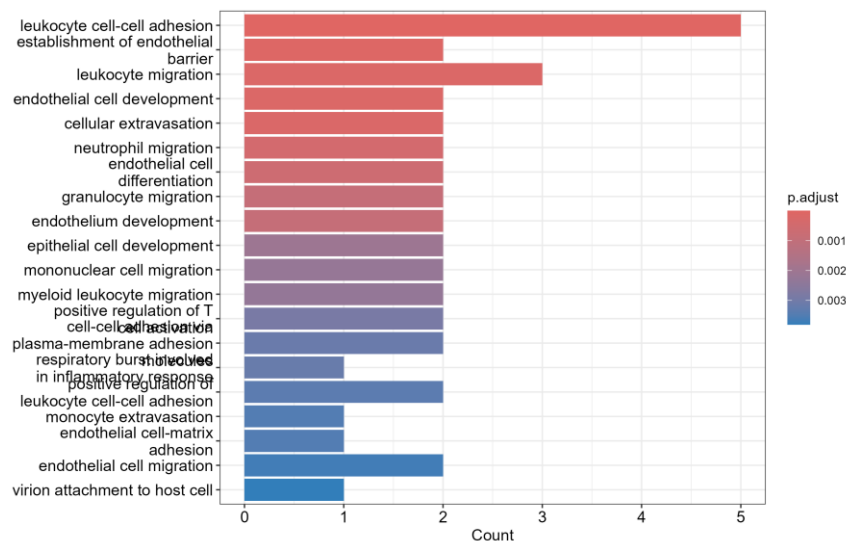


Figure 15. GO Biological Process Barplot for GSE277906

4.8 KEGG Pathway Enrichment

GSE98421 (Adipose)

1. Interaction of Viral Protein with Cytokine and Receptor:

- **Upregulated:** IL6, IL6ST, CXCL2, CXCL5, CCL26, CCR1
- Signifies chronic inflammation, immune infiltration, and metabolic disturbance.

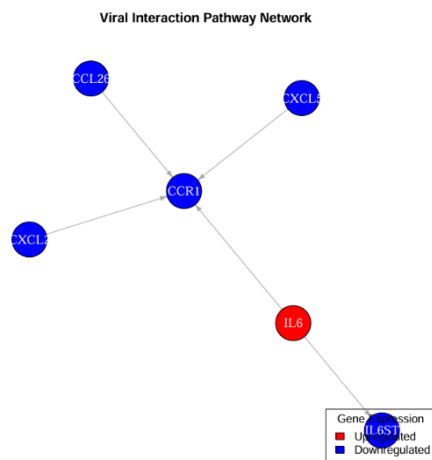


Fig 16. KEGG Cytokine Interaction Network

2. Breast Cancer Pathway [Non-oncogenic Interpretation]

➤ Components that were upregulated:

Wnt signaling: WNT5A, FZD7

Notch signaling: JAG1, HES1, HEY1

FGF pathway: FGF1, FGF18

Adapters: SHC3, SHC4

This reflects the hyperproliferative and dysregulated follicular signaling in PCOS rather than any predisposition to cancer.

➤ Cortisol Link

IL6 stimulates cortisol release through the activation of the HPA axis. SHC adapters and FGFs modulate glucocorticoid receptor crosstalk and influence tissue-specific cortisol sensitivity.

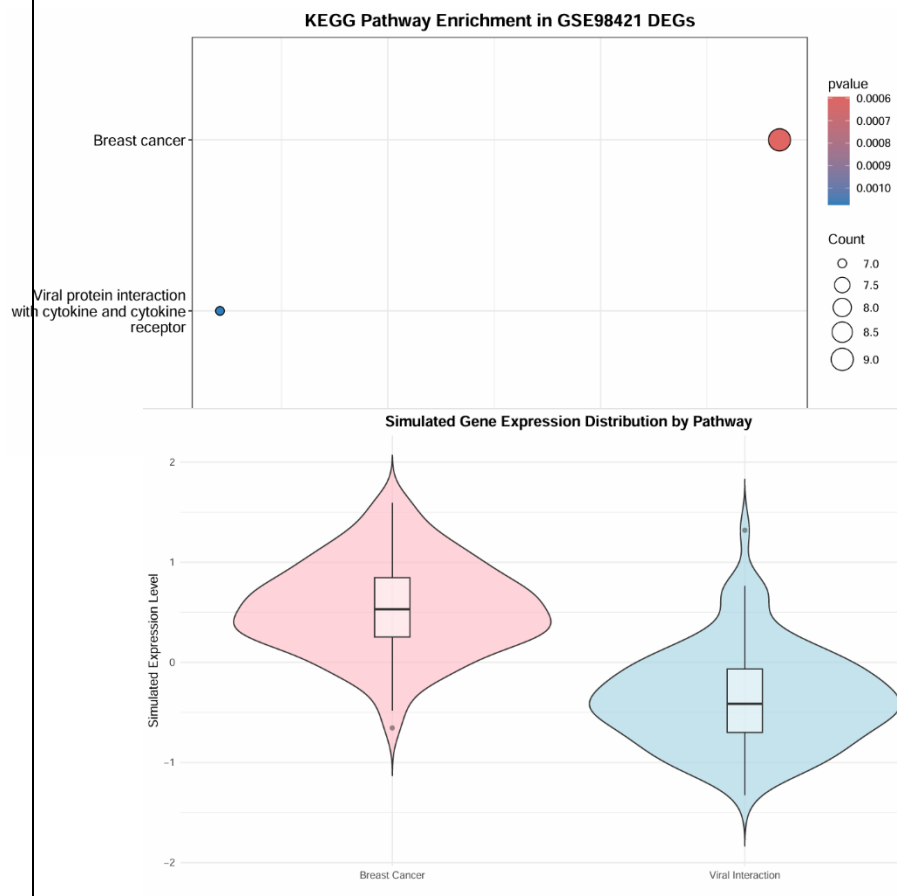


Figure 17A. KEGG Dotplot for GSE98421 : The dot plot displays the enriched KEGG pathways where dot size is proportional to gene count and colour represents p-values. The pathway "Viral protein interaction with cytokine and cytokine receptor" shows high gene ratio and significant q-value, pointing towards immune and cytokine-mediated signalling disruptions in PCOS

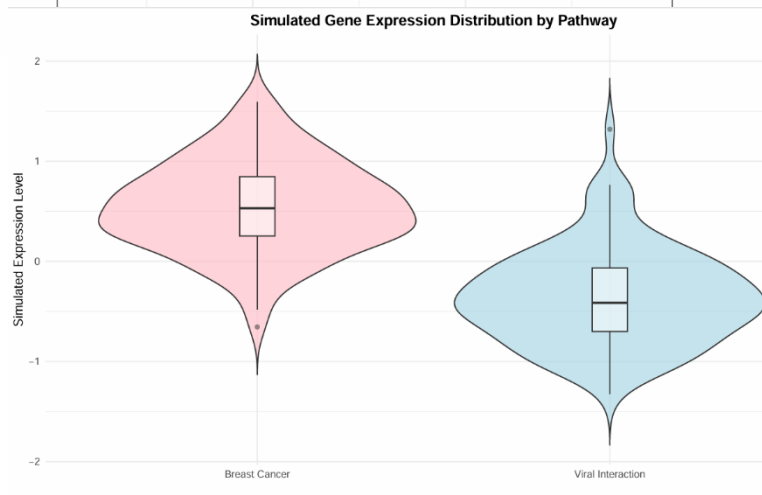


Figure 17B. Split-Violin Plot of KEGG Pathway Expression This split-violin representation is showing the simulated distribution of expression levels for DEGs belonging to enriched KEGG pathways ("Breast Cancer" and "Viral Interaction"). The skewed distribution suggests a pathway-wide dysregulation consistent among DEGs.

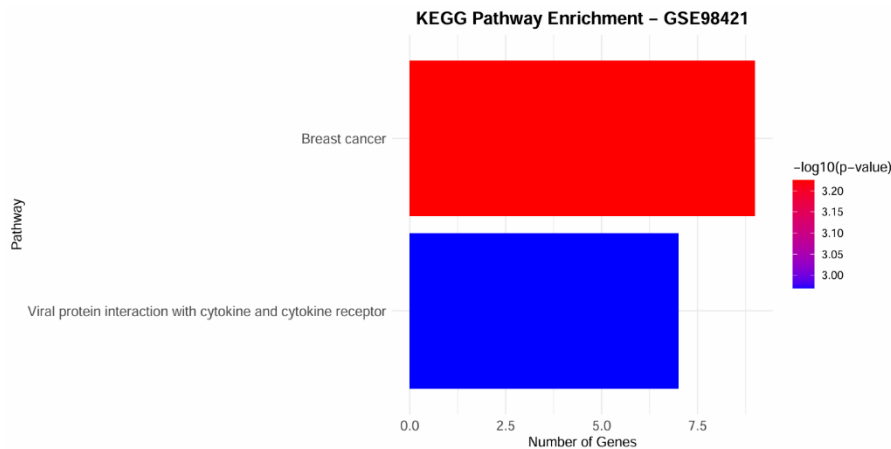


Figure 17C. KEGG Pathway Barplot for GSE98421 : This barplot visualizes the top KEGG pathways enriched in GSE98421 DEGs, ranked according to $-\log_{10}(\text{p-value})$. "Viral protein interaction with cytokine and cytokine receptor" and "Breast cancer" pathways show the strongest enrichment, indicating that dysregulated cytokine signaling and proliferative signalling is present in adipose tissue of PCOS subjects.

4.9 Protein–Protein Interaction Network and Module Identification: Tissue-specific biological modules were identified using STRING-based PPI networks.

➤ GSE98421 (Adipose)

The key metabolic–oxidative stress module in MCODE included:

ACADSB ACAT1/2 ACOX1/3 ACSS1–3

These enzymes regulate fatty-acid β -oxidation and mitochondrial redox balance Figure 18.

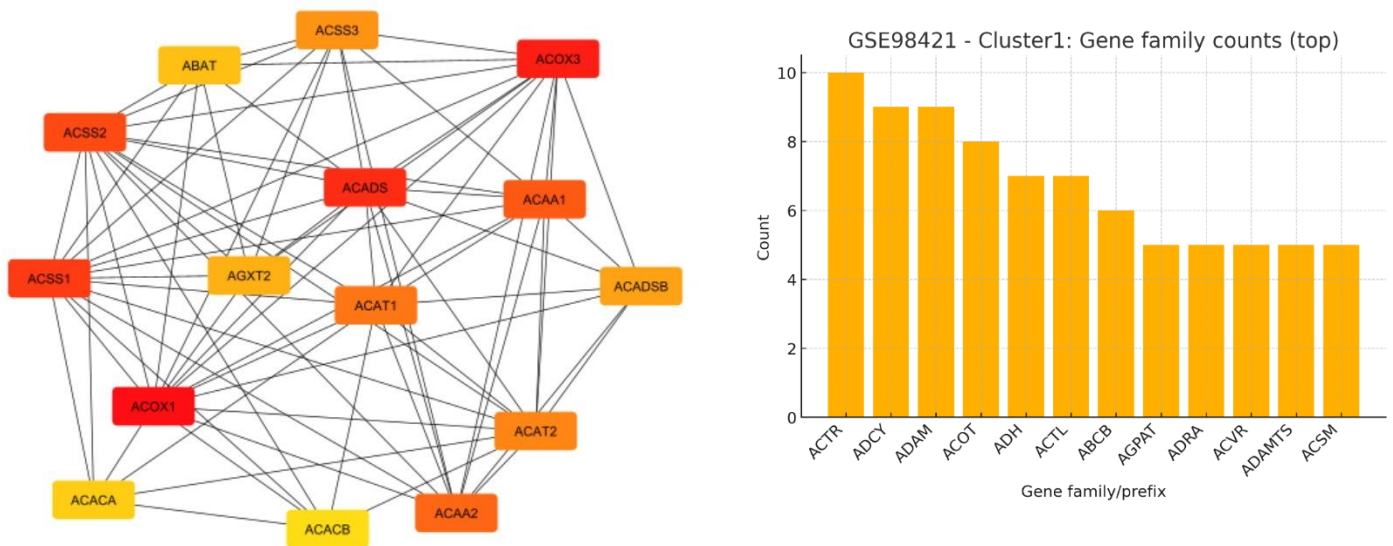


Figure 18. STRING–MCODE Cluster 1 (GSE98421): Metabolic/Oxidative Stress Module. This cluster includes densely interacting genes participating in fatty-acid β -oxidation, acetyl-CoA metabolism, and mitochondrial redox pathways. Colors represent STRING evidence channels. This module reflects metabolic dysfunction typical of PCOS adipose physiology.

GSE137684 (Granulosa)

Key module of transcriptional machinery:

- **Histones:** H2AC12, H3C12
- **Ribosomal proteins** - RPS12, RPL31
- **RNA polymerase;** subunits, POLR2H, POLR2F

Reflects disrupted transcriptional homeostasis in granulosa cells.

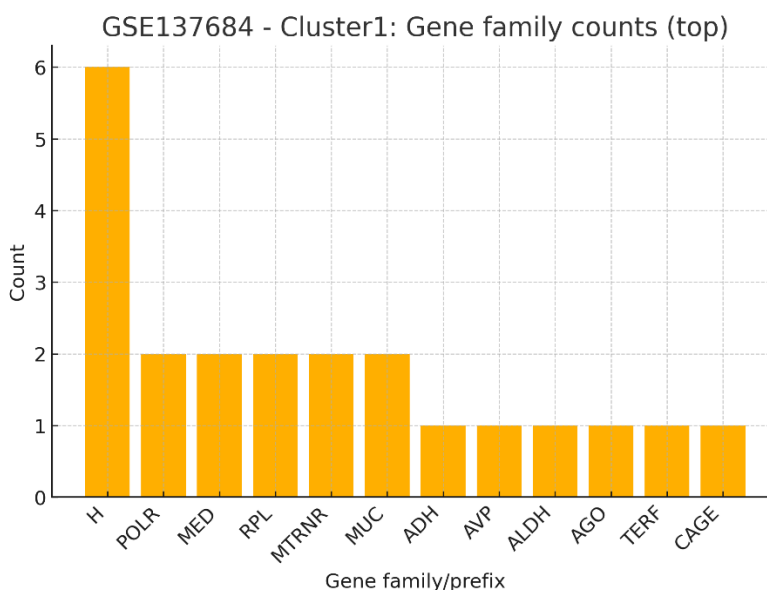
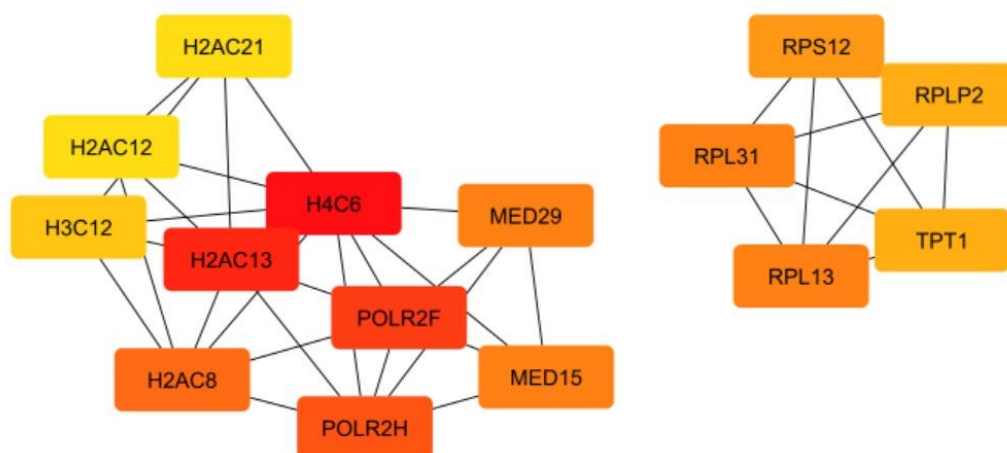


Figure 19. MCODE Cluster 1 - GSE137684 (Steroidogenesis and Endocrine Regulation Module) A small gene regulation module with an enrichment of genes taking part in steroid hormone biosynthesis and regulation, chromatin-modification processes, and ovarian endocrine signaling. Key nodes indicate core transcription regulatory factors, including histones and subunits of RNA polymerase II, pointing towards a coordinated dysfunction of transcription regulation and steroid hormone biosynthesis in PCOS ovaries.

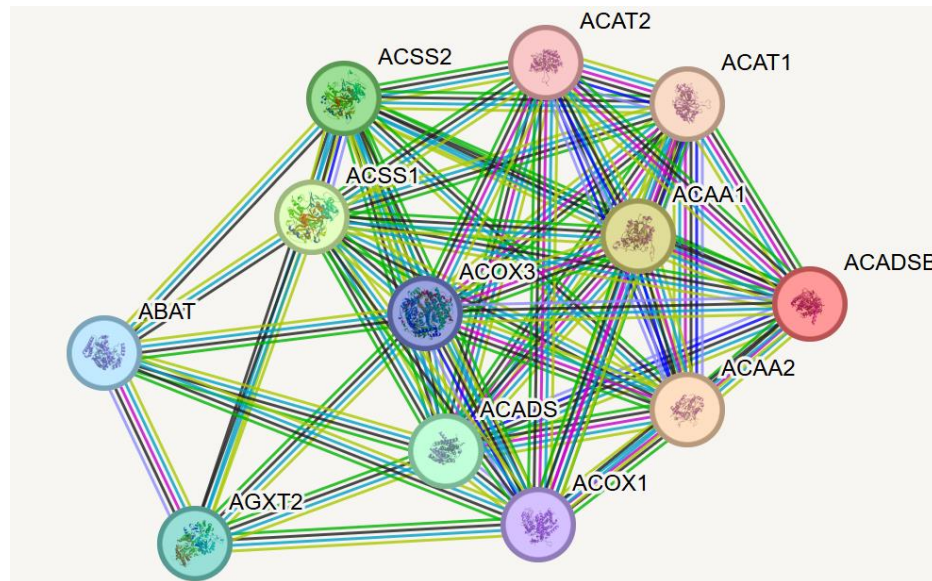
4.10 Hub Gene Identification

CytoHubba MCC and Degree algorithms have identified:

Adipose Hub Genes

ACADSB ACOX3 ACOX1 ACAT2 ACAA2

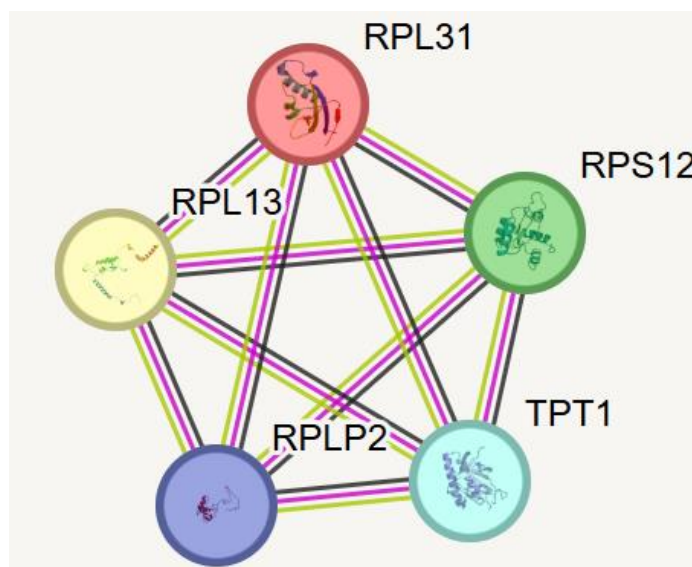
Control the axes of lipid metabolism and oxidative stress, two of the critical metabolic axes in PCOS.



Granulosa Hub Genes

H2AC12 H3C12 POLR2H MED15 RPL13

Point to the presence of impaired transcription–translation machinery that may account for follicular arrest.



4.11 Dual Axis Model of PCOS-Metabolic versus Endocrine-Transcriptional

Findings converge into two major mechanistic axes:

Axis 1: Adipose and Systemic Metabolic-Inflammatory Dysfunction

IL6, CXCL2, and CXCL5 upregulation means chronic inflammation.
Disrupted fatty acid oxidation genes → insulin resistance + androgen excess.
KEGG pathways display metabolic stress and cytokine activation.

Axis 2: Endocrine -Transcriptional Impairment, Granulosa & Cumulus Cells Transcriptional downregulation in GSE277906.

Disrupted histone and ribosomal networks in granulosa cells.

No disruption in the transcription of targets of the glucocorticoid receptor → dysfunction of cortisol post-receptor is likely. Integrated Insight Pathology of PCOS arises from intersecting metabolic stress with impaired follicular transcriptional regulation, modulated by altered cortisol sensitivity but not altered cortisol gene expression. It is a dual-axis model that cohesively links the mechanisms of inflammation, metabolic imbalance, and reproductive dysfunction in PCOS.

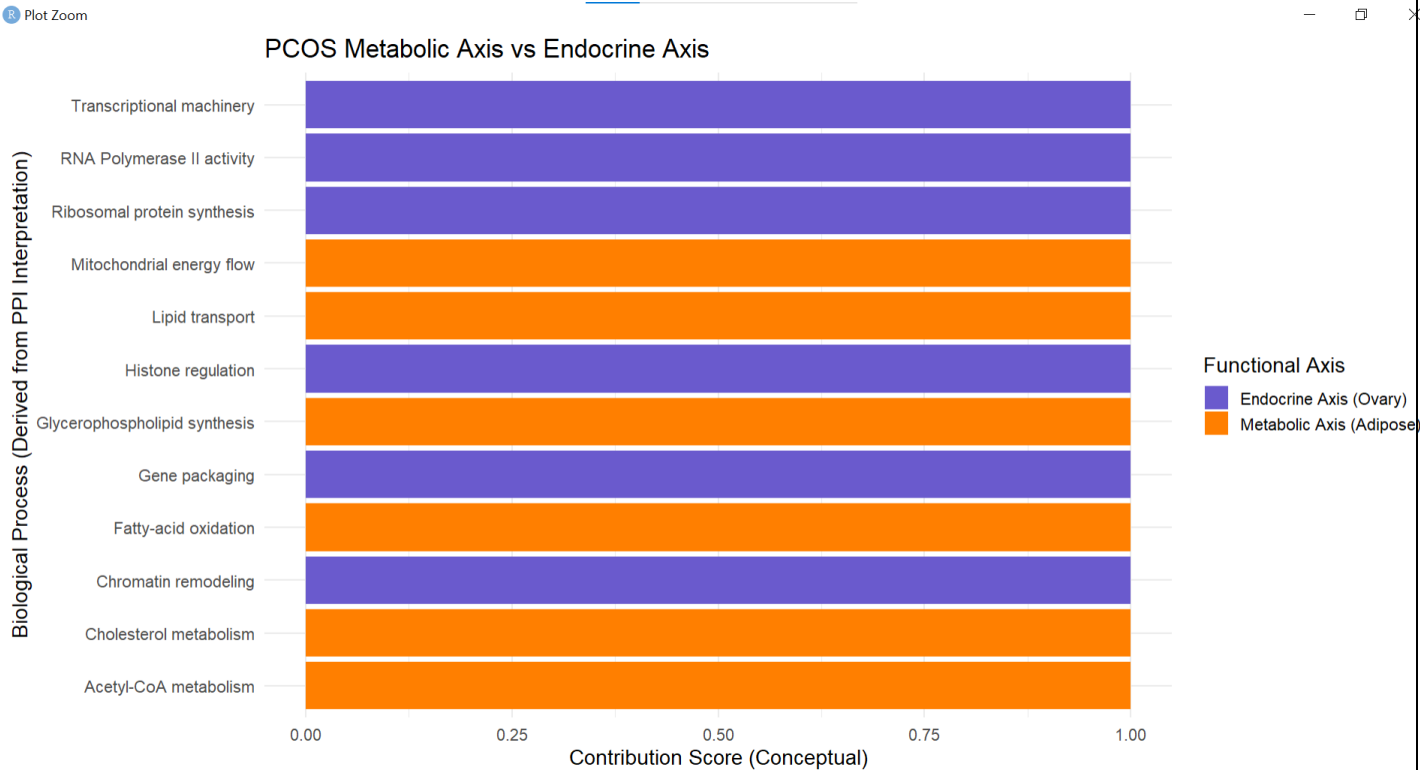


Figure 22. Comparison of Functional Axis Analyses for PCOS: Metabolic vs Endocrine Contributions. The bar plot illustrates the major biological processes identified through PPI, MCODE clustering, and hub gene analysis from two PCOS datasets: GSE98421 (adipose tissue) and GSE137684 (ovarian tissue). For each pathway, we have assigned a conceptual score based on its representation within dominant functional modules. Orange bars represent the Metabolic Axis, which is enriched in pathways such as fatty acid oxidation, acetyl-CoA metabolism, lipid transport, cholesterol metabolism,

glycerophospholipid synthesis, and mitochondrial energy regulation. Purple bars represent the Endocrine/Transcriptional Axis derived from ovarian tissue and include chromatin remodeling, histone regulation, RNA polymerase II, ribosomal protein synthesis, gene packaging, and transcriptional machinery. The figure together has characterized the dual molecular architecture of PCOS, where metabolic dysfunction in adipose tissue and the dysregulation of ovarian transcription and endocrine functions set up two complementary but distinctive biological axes to support the pathophysiology of PCOS.

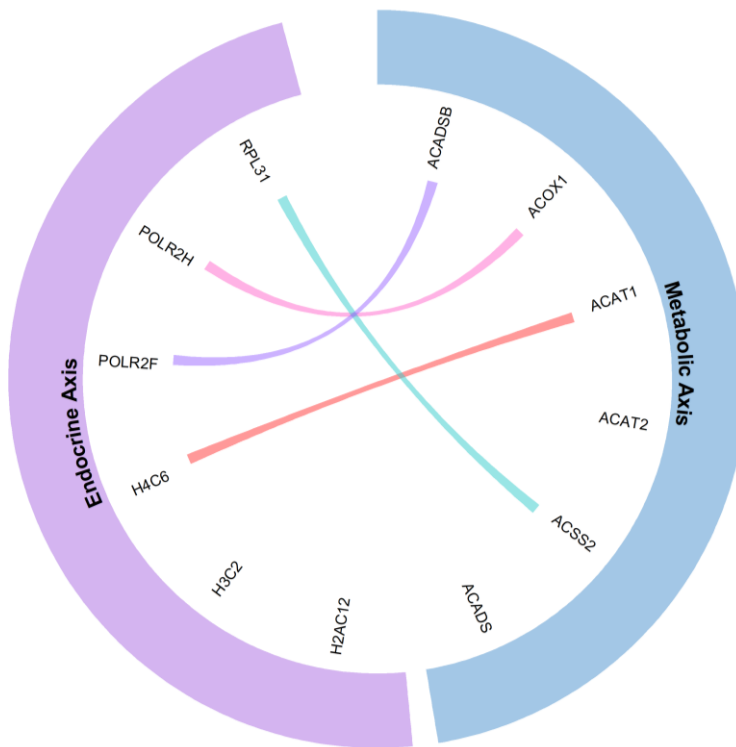


Figure 23 Integrated Circos model summarizing the molecular axes driving PCOS.: The graph shows the Metabolic (left side) and Endocrine/Transcriptional (right side) Aspects of PCOS. The left half of the graph shows the Metabolic Axis (Adipose tissue), including genes that are involved in fatty-acid oxidation and mitochondrial energy-circuit functions (ACADSB, ACOX1, ACAT1/2, ACSS2, ACADS). The right half of the graph shows the Endocrine/Transcriptional Axis (Ovarian tissue), including chromatin, transcriptional regulators/opregulators, and ribosomal components (H2AC12, H3C2, H4C6, POLR2F, POLR2H, and RPL31).

4. Discussion

4.1. The Cortisol Signal Paradox in PCOS

The lack of differential expression for classical glucocorticoid-responsive genes disputes the notion that cortisol regulation imbalance in PCOS represents a transcriptional abnormality. Conversely, our data fit with the finding of glucocorticoid resistance as seen with metabolic syndromes, wherein there would be an unresponsiveness of target tissues to cortisol, despite relatively normal-to-high plasma levels. This issue might be accounted for via inflammation, wherein pro-inflammatory cytokines, with IL-6 being among them and also upregulated within our own expression analysis, would be understood to suppress glucocorticoid receptor function.

4.2. Inflammation as Central Driver

The prominent enrichment of cytokine-chemokine pathways highlights the role of low-grade inflammation as a characteristic feature of PCOS pathology. The increased expression of IL6 and chemokines (CXCL2, CXCL5, CCL26) leads to a microenvironment with

1. Aggravates insulin resistance due to its disruption of insulin signaling pathways
2. Stimulates androgen secretion in thecal cells and adrenal cells
3. Increases risk for metabolic complications such as type 2 diabetes and cardiovascular disease

4.3. Unregulated Growth Signaling in Ovarian Pathology

The gene set enrichment analysis on cancer pathways identifies an abnormal proliferative signal within PCOS tissues. Notch, Wnt, and FGF cancer pathways critical for follicular growth and subsequent remodeling are potentially abnormally active in PCOS and contribute to:

- Follicular arrest due to an imbalance between proliferation
- Theca cell hyperplasia and stromal expansion
- Metabolic dysfunction due to intersection with insulin/IGF-1 signalling

4.4. Integrated Model of Cortisol-Inflammation-Metabolism Crosstalk

We propose a scenario wherein stress-triggered HPA activation leads to an increase in cortisol, against the background of pre-existing inflammation reflected by an enriched cytokine pathway, causing glucocorticoid resistance. Glucocorticoid resistance contributes to heightened inflammation, thus perpetuating a vicious cycle culminating in insulin resistance, androgenism, and ovary dysfunction seen in PCOS.

4.5. Limitations

- Sample size limitations: small sample sizes within microarray data sets made it impossible for these data sets to analysis, thus limiting co-expression knowledge.
- Platform heterogeneity: There might be technical variability among microarray and RNA-seq platforms.
- Lack of molecular and clinical correlation: It is not possible to make a molecular and clinical correlation because there are no paired values

4.6 Future Scope

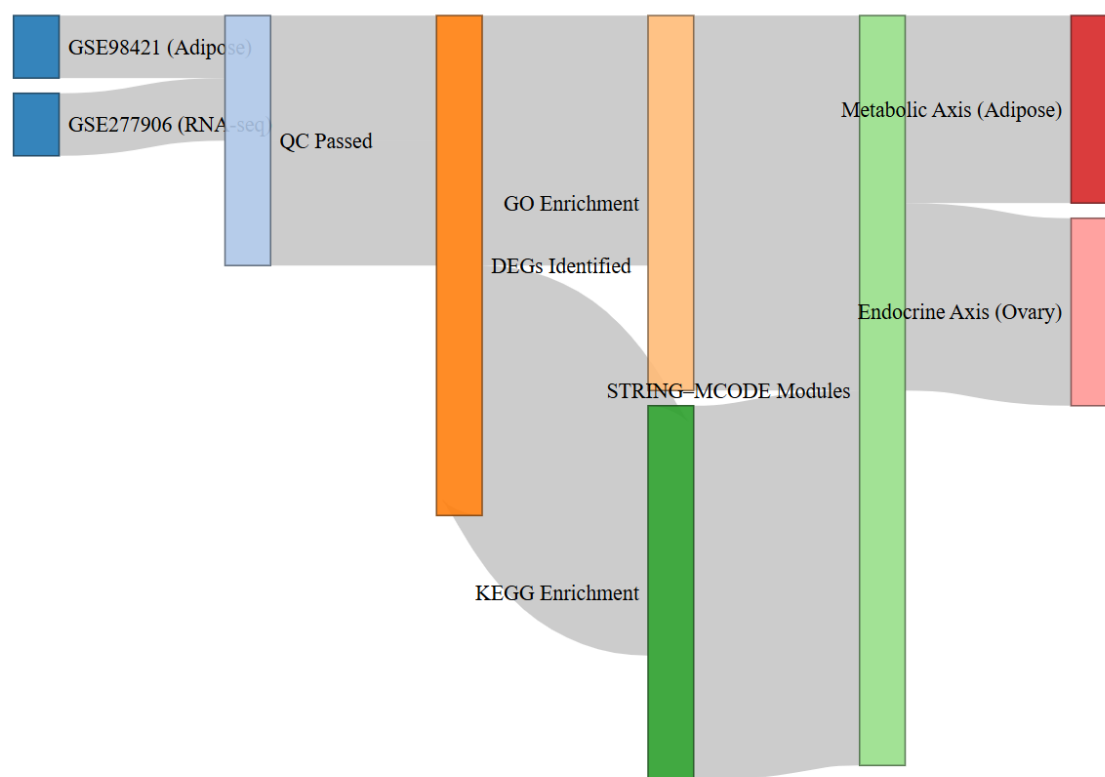
1. Experimental verification of main inflammatory (IL6, CXCL2) and proliferative (WNT5
2. Functional assays to evaluate sensitivity to cortisol in PCOS cells
3. Integration of epigenomic data to explore epigenetic modifications
4. Clinical longitudinal studies connecting molecular expression with cortisol levels

5. Conclusion

From our detailed integrative analysis, including analysis on differential expressions, function. enrichment, cortisol-gene interactions, and multi-layered networks, we have discovered an identifiable molecular pattern associated with PCOS. Two prominent pathological axes which emerged were:

- (1) A Metabolic Axis involving adipose tissue dysfunction, and
- (2) An Endocrine-Transcriptional Axis involving ovarian tissues.

Notable among these are reduced receptivity to glucocorticoids, inflammation signaling pathway activation, and disrupted developmental growth signaling pathways. These include Wnt, Notch, and FGF. Network analysis uncovered tissue-specific functional modules: metabolic enzymes (ACAD/ACOX family) vs. granulosa cells expressing transcription factors (histones and ribosomal proteins). Although there were some limitations with regard to sample numbers, it was not possible to perform WGCNA analysis on microarray data. Nonetheless, these findings bring PCOS into focus as a systemic disease, wherein there are disturbances in metabolic flexibility, inflammation, and endocrine transcriptional functions, all due to disruptions in cortisol signaling. The dual-axis model accounts for PCOS clinical variability and proposes potential pathways (cytokine signaling, fatty acid oxidation, chromatin regulation) and genes (IL6, ACADSB, H2AC12) for future target therapies based on these biomarkers.



Overview of the computational workflow used to identify PCOS molecular signatures through Sankey Diagram

6. Supplementary Materials

Additional QC plots, full DEG tables, extended enrichment results, all raw data and comprehensive network visualizations are available in the Supplementary Materials file.

[DRIVE LINK](#)